

Application Module^X System Administration

**Release 300
AX11-300**

Application Module^x

**Application Module^x
System Administration**

**Release 300
AX11-300
01/99**

Notices and Trademarks

© Copyright 1998 by Honeywell Inc.

Revision 3 – January 21, 1998

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a particular purpose and makes no express warranties except as may be stated in its written agreement with and for its customer.

In no event is Honeywell liable to anyone for any indirect, special or consequential damages. The information and specifications in this document are subject to change without notice.

Honeywell, **TotalPlant**, and Universal Station^x are U.S. registered trademarks of Honeywell Inc.

Other brand or product names are trademarks of their respective owners.

Honeywell
Industrial Automation and Control
Automation College
2820 West Kelton Lane
Phoenix, AZ 85053-3028
1-800-852-3211

About This Publication

This publication is intended to support the Application Module^X System Administrator in performing the initial configuration and subsequent administration of the Application Module^X. It is intended to supplement, but not replace, the HP-UX documentation that is available on CD-ROM.

This publication replaces publication AX11-200. It supports A^XM Release 300, and TPS system software Releases 433, 531, and 600, and later.

Table of Contents

1. INTRODUCTION	15
1.1 Overview	15
What is the Application Module ^X ?	15
Levels of System Administration	15
Scope of this manual	15
1.2 Duties and Skills of a System Administrator	16
Duties of a System Administrator	16
Additional duties	16
Prerequisite skills	17
1.3 Where to Learn More	18
HP CD-ROM manuals	18
Hewlett-Packard web site	18
Online version of the HP-UX Reference	18
Limitations of the man command	19
2. INITIAL SYSTEM SETUP	21
2.1 Overview	21
What will be covered	21
System administration strategy	21
Initial configuration of networking	22
2.2 System Administration Station Setup	23
What will be covered	23
Preparation for set_parms	23
IMPORTANT	23
Information required for set_parms	24
set_parms options	25
Connect cables	26
Configure networking	27
Set up /etc/hosts and /.rhosts files	28
2.3 A^XM Setup	29
Overview	29
Dumb terminal procedure	29
U ^X S procedure (R100)	30
U ^X S procedure (R200)	30
HP workstation procedure	31
PC procedure	32
Console Log in Procedure	34
Run set_parms on the A ^X M	34
Configure other hosts	35
Configure remote host permissions	37
Check access to the System Administration Station	39
Select AM processor type	39
End console session	40
Test networking	40

Table of Contents

2.4	Creating a New Node	41
	Overview.....	41
	What will be covered	41
	Prepare NCF backup media	42
	Installation on History Module	42
	Configure the A ^x M on the LCN	43
	Determine checkpoint volume requirements	45
	Configure checkpoint volume on the HM.....	45
	Load AM personality with null checkpoint.....	46
	Checkpoint the AM	47
	Shutdown node	47
	Reload other nodes	47
	Load the AM from the coprocessor hard drive.....	48
3.	APPLICATION MODULE^x OPERATIONS	49
3.1	Overview	49
	What will be covered	49
3.2	Bootloading the Coprocessor with HP-UX.....	50
	Two methods will be covered	50
	Cycle power	50
	WSI Restart Schematic	51
3.3	Configuring AM External Load Modules.....	52
	Overview.....	52
	Modules covered here	52
	Function of AMCL06_2.....	52
	Discussion	53
	Function of XACCES	54
	Installation on History Module	54
	Configure the NCF	55
3.4	Loading the AM Personality.....	56
	Overview.....	56
	Coprocessor Status	56
	Performing the load	57
	For more information	58
3.5	Shutting Down HP-UX.....	59
	Who can shut down HP-UX	59
	Partial vs. total.....	59
	Partial Shutdown	60
	Total shutdown	61
	For more information	61
	Using SAM for shutdown	61
	Shutdown from a US, U ^x S or GUS schematic	62
4.	BASIC SYSTEM ADMINISTRATION TASKS	63
4.1	Starting a System Administration Session	63
	Starting SAM	63
	Running SAM from the console device	64
	Running SAM from a PC or GUS using telnet.....	64

Table of Contents

4.2	Backups and Restores	65
	Backup procedure with remote DAT	65
	Backup procedure with local DAT	67
	Restore procedure with remote DAT	68
	Restore Procedure with local DAT	70
	Using Ignite-UX for System Backup and Recovery.....	71
	Restoring Updated Files	75
4.3	Adding and Removing Users.....	76
	Adding a user.....	76
	Deleting a user.....	77
4.4	Adjusting Swap Space	78
	Overview.....	78
	Recommended swap space.....	78
	Swap space procedure	79
4.5	Changing the Size of Logical Volumes	80
	Overview.....	80
	Changing logical volume sizes	80
4.6	Adding a Second Disk Drive.....	82
	Install the physical drive.....	82
	Configuration procedure	82
4.7	Adding a CD-ROM.....	83
	Install the drive.....	83
	Configuration procedure	83
	Removing the CD-ROM disk.....	84
	Inserting a CD-ROM disk.....	84
	Accessing the CD-ROM.....	84
4.8	Loading the Primary Drive	85
	Overview.....	85
	Requirements	85
	Install New Primary Drive.....	86
4.9	Upgrading the Secondary Drive	87
	Overview.....	87
	Secondary drive upgrade procedure	87
5.	INSTALLING AND UPDATING SOFTWARE	89
5.1	Overview	89
	Scope.....	89
	Software Distributor format strategy	89
	Tar-format strategy	89
5.2	Loading HP-UX.....	90
	Overview.....	90
	Requirements	90
	Installation procedure	91

Table of Contents

5.3	Copying Software to a Depot.....	97
	Overview.....	97
	Requirements	97
	Copy procedure.....	97
5.4	Installing Other Software.....	100
	Overview.....	100
	Requirements	100
	Installation procedure	101
5.5	Loading from a Tar-Format Tape.....	104
	Overview.....	104
	Installation procedure with remote DAT	104
	Installation procedure with local DAT	105

Tables

Table 2-1	Connecting A ^x M Cables.....	26
Table 2-2	Running set_parms on an HP-UX Station	27
Table 2-3	System Administration Station Network Configuration	28
Table 2-4	Setting up a Dumb Terminal.....	29
Table 2-5	Starting VT100 Emulation on an R100 U ^x S.....	30
Table 2-6	Starting VT100 Emulation on an R200 U ^x S.....	30
Table 2-7	Starting VT100 Emulation on a Hewlett-Packard Workstation	31
Table 2-8	Starting VT100 Emulation on a PC using HyperTerminal	32
Table 2-9	Starting VT100 Emulation on a PC using CRT.....	33
Table 2-10	Console Login.....	34
Table 2-11	Running set_parms on the A ^x M.....	34
Table 2-12	Identifying Other Hosts in /etc/hosts	35
Table 2-13	Configuring Remote Host Permissions.....	37
Table 2-14	Testing Access to the System Administration Station	39
Table 2-15	Ending a Console Session	40
Table 2-16	Testing Networking	40
Table 2-17	Preparing NCF Backup Media.....	42
Table 2-18	Installing External Load Modules on the HM	42
Table 2-19	Configuring the A ^x M on the LCN	43
Table 2-20	Determining Checkpoint Volume Requirements.....	45
Table 2-21	Configuring HM Checkpoint Volume	45
Table 2-22	Loading AM Personality and Null Checkpoint.....	46
Table 2-23	Checkpoint the New AM	47
Table 2-24	Shutting Down the New AM.....	47
Table 3-1	Bootloading HP-UX by Cycling Power.....	50
Table 3-2	Bootloading the Coprocessor from a Schematic	51
Table 3-3	Installing External Load Modules on the HM	54
Table 3-4	Configuring External Load Modules in the NCF	55
Table 3-5	Procedure to Load AM Personality.....	57
Table 3-6	Partial Shutdown of HP-UX	60
Table 3-7	Total Shutdown of HP-UX	61
Table 3-8	Shutting Down the Coprocessor from a Schematic.....	62
Table 4-1	Starting SAM	63
Table 4-2	A ^x M Backup with Remote DAT.....	65
Table 4-3	A ^x M Backup with Local DAT	67
Table 4-4	A ^x M Restore with Remote DAT	68
Table 4-5	A ^x M Restore with Local DAT	70
Table 4-6	Creating a Bootable System Recovery Tape	72
Table 4-7	Performing a System Recovery	72
Table 4-8	Restoring File Backups.....	75
Table 4-9	Adding a User.....	76
Table 4-10	Removing a User.....	77
Table 4-11	Recommended File System Swap	78
Table 4-12	Configuring Additional File System Swap.....	79
Table 4-13	Changing Logical Volume Size Using SAM.....	80
Table 4-14	Changing Logical Volume Size Using the Command Line	81
Table 4-15	Configuring a Second Disk Drive.....	82
Table 4-16	Configuring a CD-ROM Drive	83
Table 4-17	Accessing the A ^x M CD-ROM.....	84
Table 4-18	Installing a New Primary Drive	86
Table 4-19	Upgrading the Secondary Drive	87

Tables

Table 5-1	Loading the HP-UX base system	91
Table 5-2	Using swcopy	97
Table 5-3	Using swinstall.....	101
Table 5-4	Installing Software from a Tar-Format Tape (Remote DAT)	104
Table 5-5	Installing Software from a Tar-Format Tape (Local DAT)	105

Acronyms

A ^X M	Application Module ^X (Application Module with Extensions)
DAT	Digital Audio Tape
LAN	Local Area Network
LCN	Local Control Network
PIN	Plant Information Network
SAM	System Administration Manager
TDC	Total Distributed Control
US	Universal Station
U ^X S	Universal Station ^X (Universal Station with Extensions)
WSI	Workstation Interface
TPS	TotalPlant Solutions

References

Publication Title	Publication Number	Binder Title	Binder Number
Application Module ^x Customer Release Guide	AX04-300	Application Module ^x	3094
Application Module ^x User Guide	AX09-300	Application Module ^x	3094
Application Module ^x Troubleshooting	AX13-300	Application Module ^x	3094
Application Module ^x Service	AX13-410/510	Application Module ^x	3094
LCN System Installation	SW20-500	LCN Installation	3025

Hewlett-Packard CD-ROM Documentation

Hewlett-Packard Documentation at <http://www.docs.hp.com>

1. Introduction

1.1 Overview

What is the Application Module^X?

The Application Module^X (A^XM) is a major component of Honeywell's **TotalPlant** Open Solutions. It uses a dual-processor architecture to provide the reliability, security, and process orientation of the Application Module, augmented by a powerful coprocessor based on the Hewlett-Packard PA-RISC processor running the HP-UX operating system. The coprocessor provides an open environment for development and execution of custom control applications provided by Honeywell, by customers, and by 3rd-party sources.

Levels of System Administration

The Application Module^X can be used in different ways, each requiring a different level of System Administration support. It can function as a “black box” controller—preloaded at the factory with one or more applications that start automatically when the system is installed and powered up (or that are initiated from CL programs). In this configuration, a minimum of System Administration may be required. At the other extreme, the A^XM can function as a component of a complex, networked, multi-user development environment. Here, the System Administration tasks will be much more complex and will require considerably more expertise.

Scope of this manual

This manual is intended to supplement rather than duplicate Hewlett-Packard documentation that is available on CD-ROM and at their website. It is not intended to be a stand alone system administration manual—it should be used in conjunction with the HP documentation. This manual provides information specific to the A^XM, as well as information about basic HP-UX system administration tasks to help you get started.

1.2 Duties and Skills of a System Administrator

Duties of a System Administrator

A System Administrator is normally responsible for installing, configuring, and maintaining:

- Operating system software (HP-UX for the A^xM)

- Applications software

- Networking software

- Peripherals

- License software

Other duties include:

- Managing user accounts

- Managing and protecting the file system

- Managing system security

- Troubleshooting when strange things occur

- Training others in the use of the system

Additional duties

The System Administrator may be responsible for the administration of more than one system.

In some cases, the System Administrator may also be the Network Administrator.

Prerequisite skills

There are basic skills that you should have before attempting to administer an HP-UX system. The following list identifies some of the things that you should be able to do:

- Log in and out of your system
- Move about the HP-UX directory tree (change directories)
- Distinguish between “absolute” and “relative” path names
- Edit files using one of the HP-UX editors
- Display the contents of files using the `cat` or `more` command
- Search for text in files using the `grep` command
- Move, copy, and remove files and directories
- Use at least one of the HP-UX shells (such as `sh`, `csh`, `ksh`)
- Display manual reference pages using the `man` command
- Access CD-ROM documentation
- Use a DAT tape drive

1.3 Where to Learn More

HP CD-ROM manuals

Hewlett-Packard documentation on CD-ROM can be accessed on an A^XM, a U^XS, or on a workstation or other compatible Plant Information Network (PIN) device equipped with a CD-ROM drive. The CD-ROM documentation contains more than 200 manuals that cover HP-UX and the various applications (editors, compilers, networking software, and other tools) that are bundled with your HP-UX system. Even the most experienced UNIX System Administrator will want to refer to these manuals for information about features that are unique to HP-UX (for example, SAM—the System Administration Management tool).

Hewlett-Packard web site

Hewlett-Packard HP-UX documentation can be accessed at <http://www.docs.hp.com>. This site contains a wealth of information including white papers and Frequently Asked Questions (FAQs).

Online version of the HP-UX Reference

The online version of the *HP-UX Reference* is a powerful and fast way to access information. It is particularly useful when you are in a terminal session and need information about a command. The *HP-UX Reference* is also available in the CD-ROM documentation. The online version is often referred to as the “manual pages” and is accessed with the `man` command.

For example, if you want information about the list command `ls`, you would enter the command:

```
man  ls
```

To learn about the `man` command, enter the command:

```
man  man
```

The `man` command automatically pipes (sends its output to) the `more` command so that when there is more than one screen of information available, you use the space bar key to scroll through the information screen-by-screen.

You could also learn about the `man` command by looking it up in the CD-ROM version of the *HP-UX Reference*. It is in Vol 1: Section 1.

It is also possible to get information about certain files. For example:

```
man  hosts.equiv
```

gives you information about the contents and use of the file `hosts.equiv`.

Limitations of the man command

While the `man` command is handy for a quick online reference, it does not replace the full CD-ROM documentation set. The CD-ROM set is far more complete. Application developers and system administrators will find that the CD-ROM set is an invaluable resource.

2. Initial System Setup

2.1 Overview

What will be covered

This section covers the setup, configuration, and testing steps that you need to perform in order to prepare your system for running applications, for optionally developing applications, and for performing system administration tasks. The section is divided into two parts:

System Administration Station Setup

A^XM Setup

System administration strategy

The expert user may prefer to perform system administration tasks in the traditional way—by issuing commands from the command line and using an editor to modify system configuration files. However, we recommend using the System Administration Manager (SAM)—a tool provided by Hewlett-Packard that simplifies many common system administration tasks.

System administration tasks can be performed from a local console terminal on the A^XM, or by logging in from a remote device on the PIN. SAM will run in either environment, but it is friendlier when run from an X window. Therefore, the system administration procedures covered in this manual assume that the user will log in to the A^XM from a remote device on the PIN (for example, a U^XS or HP workstation). This device will be referred to as the System Administration Station.

If the A^XM does not have the optional local Digital Audio Tape (DAT) or other removable-media drive, certain system administration tasks that require a DAT drive, such as backups and restores, can use the DAT drive on the System Administration Station. However, installing HP-UX 10.20 on a system that currently has HP-UX 9.05 is a special case. The A^XM must have a locally connected DAT drive and console terminal to perform the installation.

Initial configuration of networking

You must use the console port on the A^XM to configure networking. After networking is set up, further system administration tasks can be performed over the network.

The A^XM console terminal can be a VT 100 compatible terminal, or a device such as a PC running VT 100 emulation. It can also be a U^XS or Hewlett-Packard workstation running Kermit communications software. The procedure in this section assumes that a serial port on the System Administration Station (or terminal) is connected to the console port on the A^XM. Refer to the *Application Module^X Service* manual for details.

Attention

The following “ground rules” apply in using the procedure tables in this manual: The user logs in as “root” when performing any procedures in the manual.

Procedures in Section 2 are to be performed in order unless otherwise specified.

When a procedure specifies the Enter key, the key is actually labeled Return on some keyboards.

2.2 System Administration Station Setup

What will be covered

This subsection covers the setup of the System Administration Station—the device from which you will perform system administration tasks on the A^XM. After initial configuration, you may also use this station to develop A^XM application programs. The station can be one of the following:

- An HP workstation that includes a DAT drive and CD-ROM drive

- A U^XS

A PC or GUS can also be used to access the A^XM using telnet, but these devices do not support A^XM application development themselves.

Preparation for set_parms

Certain network-related parameters must be configured before network operation is possible. Your A^XM system includes a script, /sbin/set_parms, which when run, prompts you for required information. The script will automatically make necessary changes to the various configuration files. If your System Administration Station has not already been set up for networking, you will have to do so. You can use set_parms if the station is running HP-UX.

IMPORTANT

Before running set_parms, you should have available all of the required information listed below for the options you are going to configure. Your Network Administrator can supply you with the information.

The System Administration Station and the A^XM (as well as all systems connected to the network) must have unique hostnames and Internet Protocol (IP) addresses. To avoid duplicate names or addresses, enter the following commands from another device on the network before you run set_parms on the A^XM or System Administration Station.

```
nslookup hostname
```

```
nslookup address
```

where *hostname* and *address* are the IP hostname and address that you plan to use for the A^XM. Repeat the commands for the System Administration Station if you are configuring it at the same time. You should get messages similar to the following if the hostnames and addresses are unique (the actual text depends on your network configuration.).

```
*** No address information is available for “hostname”
```

```
*** No hostname information is available for “address”
```

Information required for set_parms

The set_parms script allows you to set:

- Time zone and current time and date
- System hostname and Internet Protocol (IP) address
- Root password, if not already set
- Be a font client (not recommended for A^XM)

In addition, you should configure other optional networking parameters if they apply to your installation:

- Subnet mask
- Default Network Gateway
- Gateway hostname
- Gateway network (IP) address
- Berkley Internet Domain Name Server (BIND)
 - Local domain name
 - BIND server hostname
 - BIND server network (IP) address

Attention

After you start set_parms, it will inform you that you require certain networking parameters and ask you if you want to continue. If you answer “n” for no, HP-UX will shut down and halt. You can restart by cycling the power, or you can use the WSI Restart schematic (see subsection 3.2) from any US, U^XS or GUS. If you use the schematic, you do not have to cycle power, and therefore do not have to shut down the AM.

set_parms options

If you execute `set_parms` with no arguments, you will have the opportunity to configure all of the parameters listed previously in the Information required for `set_parms` procedure. This is how you should run `set_parms` the first time.

There are other options that allow you to configure only certain options. These may be run at a later time to add or change specific parameters.

<code>set_parms</code>	<code>hostname</code>	Lets you alter the unique system hostname
<code>set_parms</code>	<code>ip_address</code>	Lets you alter the network (IP) address
<code>set_parms</code>	<code>timezone</code>	Lets you set/change time zone
<code>set_parms</code>	<code>addl_netwrk</code>	Lets you set/change additional network parameters such as subnet mask, network gateway, network gateway IP address, local domain name, Domain Name System (DNS) server host name, DNS server IP address and Network Information Service (NIS) domain name.
<code>set_parms</code>	<code>font_c-s</code>	Lets you configure your workstation to be a font client or server.

Note: Changes you make using `set_parms` will take effect after rebooting the system.

Connect cables

Table 2-1 Connecting A^xM Cables

Step	Action
1	Power down the A^xM if it is not already powered down. Connect the serial communications cable from the A^xM console terminal port to the serial port on the System Administration Station. This procedure is covered in the A^xM Service manual. NOTE: On an R100 U^xS, this connection is to J2 on the WSI I/O board. On an R200 U^xS, this connection is to J1 on the WSI2 I/O board.
2	Connect the A^xM to the PIN (Ethernet). This procedure is covered in the A^xM Service manual. NOTE: The PIN may be as simple as a point-to-point connection from one A^xM to one U^xS or workstation.
3	Verify that the Shutdown switch on the left edge of the A^xM WSI2 board is toggled to the right position (RUN).
4	Power up the A^xM and the System Administration Station.

Configure networking

If your System Administration Station is new and has not been configured for networking, do so now. If the station is running HP-UX, you can do this by using the procedure in Table 2-2 to run `set_parms`. If the station is not on HP-UX, refer to the documentation for the station and set up the following:

Time zone and current time and date

System hostname and Internet Protocol (IP) address

Root password, and, if required,

Gateway and/or name server configuration

Table 2-2 Running `set_parms` on an HP-UX Station

Step	Action
1	Log onto the System Administration Station as root. If this is the first time that you have booted the system, it will automatically start the <code>set_parms</code> procedure. If not, and if the station needs to be set up for networking, enter the command: <code>set_parms</code>
2	Respond to the prompts and supply the required information.
3	Enter the command: <code>reboot -q</code>
4	After the reboot has completed, log in as root and enter the root password.
5	Use the following commands to test the network configuration: <code>ifconfig</code> <code>lan0</code> Shows Internet address <code>hostname</code> Shows Internet hostname <code>netstat -r</code> Shows routing tables

Set up /etc/hosts and /.rhosts files

The following procedure sets up two configuration files in the System Administration Station. In the /etc/hosts file, it adds the Internet address, hostname, and alias names of the A^xM. In the /.rhosts file, it adds the full hostname of the A^xM, which enables the A^xM to use the DAT drive on the System Administration Station (it allows a remote shell without password for root).

Table 2-3 System Administration Station Network Configuration

Step	Action
1	Skip this step if your network uses a name server. Your Network Administrator can tell you whether or not your network uses a name server. Using the editor of your choice, add the following line to the file /etc/hosts: <pre>address hostname alias1 alias2</pre> where address is the Internet (IP) address of the A^xM, and hostname is the full Internet (IP) hostname of the A^xM. This can be followed by one or more alias names by which the host can also be known. For example: <pre>163.145.190.45 axm1.iac.honeywell.com axm1</pre>
2	Using the editor, add the following line to the file /.rhosts (you may be creating this file): <pre>hostname</pre> where hostname is the full internet (IP) hostname (not an alias) of the A^xM. For example: <pre>axm1.iac.honeywell.com</pre> NOTE: If your A^xM has a local DAT drive, and if you will not need to access the DAT drive on the system administration station, you may skip this step. If you are not sure, we recommend that you perform the step.

2.3 A^XM Setup

Overview

This subsection covers the configuration of the A^XM. This involves the following steps:

Set up the System Administration Station for VT-100 emulation on its serial port.

Log into the A^XM on the console port and set up basic configuration (uses the set_parms script).

Configure the Internet (IP) names and addresses of the System Administration Station and any other stations with which the A^XM will communicate (uses SAM to modify the file /etc/hosts).

Configure the services that remote hosts will be allowed to perform when they log into the A^XM (uses SAM to modify the file /var/adm/inetd.sec).

Configure the A^XM personality load based on the LCN processor type.

Attention

Do not load the AM node personality until you have completed the following A^XM setup procedures. You must set up networking and reboot HP-UX before loading the AM. If you follow the procedures in Section 2 in the order given, you should not have any difficulties.

Dumb terminal procedure

Use the following procedure if you are using a “dumb terminal” as the A^XM console terminal.

Table 2-4 Setting up a Dumb Terminal

Step	Action
1	Refer to the terminal documentation and set up the terminal for VT100 emulation.
2	Set the following communication parameters: 9600 baud 8 data bits, 1 stop bit, no parity
3	Go to the Console Login Procedure (Table 2-9)

U^XS procedure (R100)

Use the following procedure if you are using an R100 U^XS to serve as the A^XM console. The serial connection is covered in the *A^XM Service* manual. The printer serial port on the U^XS is used.

Table 2-5 Starting VT100 Emulation on an R100 U^XS

Step	Action
1	Enter the command: <code>kermit -l /dev/lp0 -b 9600 -c</code> (l is the letter ell) NOTE: If you do not have the file /dev/lp0, you can create it with the following command: <code>mknod /dev/lp0 c 1 0x050004</code>
2	Go to Table 2-10, Console Login.

U^XS procedure (R200)

Use the following procedure if you are using an R200 U^XS to serve as the A^XM console. The serial connection is covered in the *A^XM Service* manual. The console serial port on the U^XS is used. Before you can run Kermit on the console port of the U^XS, you must stop the console process. When you have finished using Kermit, you should restart the console process (see Table 2-15).

Table 2-6 Starting VT100 Emulation on an R200 U^XS

Step	Action
1	First find the Process Identification Number (PID) of the console process. Enter the command: <code>ps -ef grep console</code>
2	You should see a line of the form: <code>root nnn...../etc/getty -h console console</code> nnn is the PID—record it here _____
3	Enter the command: <code>kill -STOP nnn</code> using the PID for nnn.
4	Invoke Kermit with the command: <code>kermit -l /dev/console -b 9600 -c</code> (l is the letter ell)
5	Go to Table 2-10, Console Login NOTE: After you have completed the configuration using Kermit, you will be instructed in Table 2-15, Ending a Console Session, to enter a command that will restore the console process on this U ^X S.

HP workstation procedure

Use the following procedure if you are using the serial port on a Hewlett-Packard workstation as the A^XM console.

Table 2-7 Starting VT100 Emulation on a Hewlett-Packard Workstation

Step	Action
1	Enter the command: kermit
2	Enter the command: set line /dev/tty00
3	Enter the command: set baud 9600
4	Enter the command: log session pathname where pathname is a file path of your choice for a session log file (for example, /tmp/session.log) NOTE: Do not skip this step, even if you don't need a log. On the 712/60, Kermit has a tendency to hang if the session is not logged.
5	Enter the command: connect
6	Go to Table 2-10, Console Login.

PC procedure

Use the following procedure if you are using a PC operating in the Windows environment as the A^XM console terminal. HyperTerminal, which is the terminal emulation software that comes with Windows, does not always produce legible screen output when used with HP-UX 10.20.

If you are reinstalling HP-UX or booting from your recovery tape, you will have better results using CRT, a terminal emulation software program available via the World Wide Web at <ftp://ftp.vandyke.com/pub/CRT/>. The installation program issues an escape sequence to the terminal, in an attempt to identify it. HyperTerminal and Terminal.exe do not respond properly to the sequence. The CRT software does respond properly, and produces legible output when the font is set to Courier 10 pt.

Table 2-8 Starting VT100 Emulation on a PC using HyperTerminal

Step	Action
1	Double-click the Accessories icon
2	Double-click the HyperTerminal icon
3	Pull down the File menu and select Properties Select the Settings tab and verify the following options are selected: Function, arrow, and ctrl keys act as: Terminal keys Emulation: Auto detect Backscroll buffer lines: 500 Select OK
4	Pull down the File menu and select Properties Select the Connect To tab and select COM 1 or COM2 Click the Configure button Make the following selections from the Port Settings window: Bits per second: 9600 Data bits: 8 Parity: None Stop bits: 1 Flow Control: None Select OK
5	Maximize the window
6	If you wish to save your configuration settings, pull down the File menu, select Save As, enter a filename, and Select OK. The next time you enter the terminal emulation program, pull down the File menu and select Open. This will automatically restore your settings.

Table 2-9 Starting VT100 Emulation on a PC using CRT

Step	Action
1	Select CRT as your terminal emulation software
2	From the Connect window, select the Quick Connect tab and verify that the following options are selected: Protocol: serial Port: COM1 or COM2 Baud rate: 9600 Data bits: 8 Parity: None Stop bits: 1 Check the Save session box Select OK
3	Pull down the Options menu and select Session Preferences. Select the Display tab . Click the Font button and choose Courier 10pt as the font. Select OK.
4	Maximize the window.

Console Log in Procedure

Table 2-10 Console Login

Step	Action
1	If the A ^x M is not powered on, turn power on now (you can watch the bootup process). NOTE: Be sure that the toggle switch on the left side of the WSI2 board is in the right (RUN) position.
2	Press Enter—you should receive the login prompt from the A ^x M.
3	Log in as root (no password will be required).
4	You will see the line: TERM= Enter the following: VT100 and press Enter

Run set_parms on the A^xM**Table 2-11 Running set_parms on the A^xM**

Step	Action
1	Refer to subsection 2.2 of this manual and note the information that will be required to run set_parms. Have this information available.
2	Enter the command: set_parms
3	Respond to the prompts and enter the required information.
4	When set_parms is done, enter the command: <code>reboot -q</code> NOTE: Do not skip this step—the networking configuration will not take effect unless the system is rebooted, and subsequent steps require it to be performed.
5	Log in as root and enter the root password.
6	Use the following commands to test the network configuration: <code>ifconfig lan0</code> Shows Internet address <code>hostname</code> Shows Internet hostname

Configure other hosts

In this procedure, you will use SAM to modify the file `/etc/hosts`, entering in the Internet (IP) addresses, names, and alias names of other stations on the network with which the A^XM will communicate.

Note: If your network uses a name server, this procedure is not necessary. Have your network administrator configure the A^XM and the System Administration Station in the server, and skip to the next procedure, “Procedure to Configure Remote Host Permissions.”

Table 2-12 Identifying Other Hosts in `/etc/hosts`

Step	Action
1	Enter the command: <code>sam</code> NOTE: To select a menu item in the following steps, use the arrow keys to highlight the item, then press Enter to select.
2	A screen appears showing information about how to use the keyboard to navigate within SAM windows. Press Enter to continue.
3	Select Networking and Communications.
4	Select Internet Addresses.
5	Press the Tab key to go to the menu bar.
6	Use the arrow key to select Actions, and then press Enter.
7	Select Add.
8	If you are using BIND (DNS), you will get an error message here (see NOTE in the paragraph before this table. If this occurs, press the Tab key to go to NO, press Enter, and go to step 13.
9	Enter the System Administration Station Internet address, hostname, and, if desired, alias names.
10	Use the Tab key to go to OK and then press Enter.
11	Repeat steps 5-10 for each additional network device with which the A ^X M will communicate.
12	Use the Tab key to go to OK and then press Enter.

Table 2-12 Identifying Other Hosts in /etc/hosts

Step	Action
13	Use the Tab key to go to the menu bar and then press Enter to select the File menu.
14	Select Exit.
15	Use the Tab key to go to the menu bar and then press Enter to select the File menu again.
16	Select Exit SAM.

Configure remote host permissions

The file `/var/adm/inetd.sec` permits you to allow or deny specific services to individual users or groups of users. Workstations are normally shipped with all of the permissions set to “allow”—minimum security. The A^XM, however, is shipped with all permissions set to “deny”—maximum security. For those remote users and services that you wish to grant access to, you must change the permissions to “allow”. In this procedure, you use SAM to configure the file `/var/adm/inetd.sec` in order to set these permissions.

Table 2-13 Configuring Remote Host Permissions

Step	Action
1	Enter the command: <code>sam</code>
2	A screen appears showing information about how to use the keyboard to navigate within SAM windows. Press Enter to continue.
3	Select Networking/Communications.
4	Select System Access.
5	Select Internet Services.
6	Use the arrow key to position the cursor at telnet Remote Login and press Enter to highlight the line.
7	Use the Tab key to go to the menu bar.
8	Use the arrow key to select Actions and then press Enter.
9	Select Modify.
10	Use arrow key to select desired permission—Honeywell recommends Selected-Allowed—press Enter.
11	For System and/or Network, enter the System Administration Station hostname (full hostname, not an alias).
12	Use the Tab key to go to Add and then press Enter (adds the hostname and permission to the list).
13	Repeat Steps 10-12 for each additional host that needs to remotely login to the A ^X M.

Table 2-13 Configuring Remote Host Permissions

Step	Action
14	Use the Tab key to go to OK and then press Enter.
15	Repeat steps 6-14 for each of the following services: shell, login, ftp, exec, and echo
16	Use the Tab key to go to the menu bar and then press Enter to select the File menu.
17	Select Exit.
18	Use the Tab key to go to the menu bar and then press Enter to select the File menu again.
19	Select Exit SAM.

Check access to the System Administration Station

If a procedure requires the A^XM to have a local DAT drive, and it does not have one, the A^XM can access the DAT drive of the System Administration Station. To do this, the backup, restore, and tar utilities will do a remote shell (remsh) to the System Administration Station. In order to work properly, the station must not prompt for a password. This is accomplished by entering the full A^XM hostname in the file `/.rhosts` in the System Administration Station, which you performed in subsection 2.2. The following procedure tests that this setup is correct.

Table 2-14 Testing Access to the System Administration Station

Step	Action
1	If your A ^X M has a local DAT drive and you did not enter the A ^X M hostname in the <code>/.rhosts</code> file in the System Administration Station while setting up the station in subsection 2.2, skip this procedure and go to the next topic, “Select AM processor type.”
2	You are still logged on the A ^X M as root. Enter the command: <code>rlogin station_name</code> where <code>station_name</code> is the Internet (IP) hostname for the System Administration Station.
3	If the login is successful, you will get a command line prompt from the remote without having to enter a password.
4	Enter the command: <code>exit</code> to break the connection.

Select AM processor type

If the processor on the AM-side of your A^XM is a K2LCN or HMPU, skip to the procedure “End console session.” If your processor is a K4LCN (applies to R500 or later), enter the following two commands to convert the personality type to K4LCN (the second command is typed on one line with a space after “axmk”):

```
rm -f /etc/opt/TDC_Open/common/personality.config
ln -s
/opt/TDC_Open/common/newconfig/personality.config.axmk
/etc/opt/TDC_Open/common/personality.config
```

If at a later time you need to convert back from a K4LCN to an HMPU or K2LCN, enter the following two commands:

```
rm -f /etc/opt/TDC_Open/common/personality.config
ln -s
/opt/TDC_Open/common/newconfig/personality.config.axm
/etc/opt/TDC_Open/common/personality.config
```

End console session**Table 2-15 Ending a Console Session**

Step	Action
1	Reboot the A ^x M by entering the command: reboot -q
2	If you were using Kermit, disconnect the Kermit connection by pressing: [CTRL] and [N] together, and then [c] and then exit Kermit by entering the command: exit
3	If you were using an R200 U ^x S as the console terminal, enter the following command in order to restore the console process on the U ^x S: kill -CONT nnn using the PID for nnn that you recorded when performing the steps in Table 2-6.

Test networking

This completes the networking configuration steps for the System Administration Station and the A^xM. The following procedure performs some simple checks to ensure that networking works correctly.

Table 2-16 Testing Networking

Step	Action
1	Go to the System Administration Station and log on. Enter the command: nslookup axmname where axmname is the IP hostname of the A ^x M. This command should return the IP hostname and address of the A ^x M.
2	Enter the command: nslookup axmaddress where axmaddress is the IP address of the A ^x M. This command should also return the IP hostname and address of the A ^x M.
3	Enter the command: /usr/sbin/ping axmname where axmname is the IP hostname of the A ^x M. This should result in a continuous series of lines similar to: 64 bytes from axmaddress : icmp_seq=1 time=2. ms 64 bytes from axmaddress : icmp_seq=2 time=2. ms 64 bytes from axmaddress : icmp_seq=3 time=2. ms 64 bytes from axmaddress : icmp_seq=4 time=2. ms where axmaddress is the IP address of the A ^x M. To stop the test, press [CTRL] and [c] together.

2.4 Creating a New Node

Overview

If your A^xM is a new LCN node (as opposed to an existing AM that is being upgraded to an A^xM), there are some special procedures that you must follow to configure the node on the LCN and create a checkpoint on the HM. If your A^xM is an upgrade of an existing AM, you can go directly to subsection 3.3 and follow the procedures to configure external load modules and load the AM personality.

What will be covered

The procedures that will be covered here are:

- Install NCF backup media

- Check external load modules on the HM and install, if necessary

- Configure the A^xM node on the LCN

- Allocate user memory

- Configure external load modules

- Load standard AM personality from the HM along with a null checkpoint from floppy or cartridge

- Checkpoint the AM

- Shut down the AM

- Load AM personality from the coprocessor hard drive

Attention

Perform the following procedures in sequence.

Prepare NCF backup media

Table 2-17 Preparing NCF Backup Media

Step	Action
1	At a US, U ^x S or GUS, insert a current NCF backup cartridge or floppy (must have a volume/directory &ASY).
2	From the Engineering Main Menu, select <code>SUPPORT UTILITIES</code> .
3	Select <code>MODIFY VOLUME PATHS</code> .
4	Set the NCF Backup Path to the NCF backup media pathname: \$Fx>&ASY> (x = drive number)

Attention

In the next procedure, you will install AMCL06_2 on the History Module. There are different versions of AMCL06 for different LCN releases. These versions are identified as AMCL06, AMCL06_1, AMCL06_2, and so forth. All the LCN releases that are valid for A^xM Release 300 require AMCL06_2.

Installation on History Module

Table 2-18 Installing External Load Modules on the HM

Step	Action
1	Using the Command Processor of the Engineering or Universal personality, determine if the directories &CUS and &CLX exist on the History Module: LSV NET (lists volumes on the network)
2	If the directories exist, verify that the directory &CUS contains the files AMCL06_2.LO and XACCES.LO, and that the directory &CLX contains the file AMCL06_2.SF: LS NET>&CUS LS NET>&CLX
3	If either or both directories do not exist, use the Create Directory command to create the missing directory or directories on the History Module: CD NET>vol>&CUS (vol is an existing volume on the HM) CD NET>vol>&CLX
4	If you had to create the directories or if they did not contain the required files, copy the files from the Application Module Personality cartridge (&C3)(x = the removable media drive number): CP \$Fx>&CUS>AMCL06_2.LO NET>CUS>AMCL06_2.LO CP \$Fx>&CUS>XACCES.LO NET>CUS>XACCES.LO CP \$Fx>&CLX>AMCL06_2.SF NET>CLX>AMCL06_2.SF

Configure the A^xM on the LCNTable 2-19 Configuring the A^xM on the LCN

Step	Action
1	From the Engineering Personality Main Menu, select LCN NODES .
2	An LCN Node Configuration display will appear with a target for each node possible on the LCN. Verify that the configuration is in the ON-LINE mode in the upper right corner of the display. Select the target for the node number of the A ^x M that you wish to configure (you may need to use the Page Forward key to display the target for the desired A ^x M).
3	Because you are configuring a new node, a screen will appear for you to select the desired node type. Select APPLICATION MODULE .
4	NOTE: In the steps that follow, do not press Enter until instructed.
5	Leave REDUNDANT MEMBER NODE ID at 0, and leave STARTUP MODE at COLD.
6	Move the cursor to the box for entry of units to be assigned to the A ^x M.
7	Enter the units separated by spaces.
8	Page forward to Page 2 of the configuration screen and enter the user memory allocation information. We suggest the following: # BACKGROUND CL TASKS—10 CONCURRENT DATA ACCESSES FROM BACKGROUND CL—4 BACKGROUND TASK STACK SIZE—15000 CVB SIZE FOR FAST & SLOW POINT PROCESSORS—(use default of 2000 or see AM Implementation Guidelines for information on how to calculate) REDUNDANCY BUFFER INCREASE—0 NCLUDE INTERNET POINT PROCESSOR—select YES if this A^xM will access points on another LCN through a Network Gateway If you entered YES above, you will be prompted for CVB SIZE FOR IPP—(use default of 2000 or see AM Implementation Guidelines for information on how to calculate) USER MEMORY RESERVED—0
9	Page forward to Page 3.

Table 2-19 Configuring the A^xM on the LCN

Step	Action
10	In the first box in the “NAME” column, enter AMCL06_2 (this module is required).
11	Move to the adjacent box in the “PERS” column and add the personality type AM0
12	If you are going to configure XACCES, enter its name XACCES and its personality AM0 in the next pair of boxes. NOTE: Installing XACCES has significant security implications—refer to subsection 3.3 in this manual and to Section 2 of the Application Module ^x User Guide for more information.
13	If you are going to require other external load modules, you may configure them in the same manner at this time (the standard external load modules are covered in appendices C-I in the CL/AM Reference Manual).
14	When you have completed entering the desired modules, and the additional module memory if required, press Enter.
15	An advisory message may appear: “TO CHANGE AN EXISTING NODE, IT MUST BE DELETED AND THEN REBUILT”—press CANCEL to clear it.
16	Press [CTL][F1] to check the new NCF, and press [CTL][F2] to install the new NCF.
17	Return to the Engineering Personality Main Menu. A message will appear at the bottom of the display indicating the successful completion of the NCF installation.

Attention

The remaining procedures in this section are required only if:

 You are adding a new node, or

 You are upgrading an existing AM to an A^xM **and** you have increased processor memory size (for example, from a K2LCN-4 to a K2LCN-8)

If you are upgrading an existing AM and are not changing the memory size, go to subsection 3.4 of this publication, “Loading the AM Personality,” read the material there, and perform the procedure to load the AM.

Determine checkpoint volume requirements**Table 2-20 Determining Checkpoint Volume Requirements**

Step	Action
1	In this step, you will determine the HM node pair that has the AM checkpoints. Using the Command Processor, enter the command: LSV NET Scan the result and find the volume: &5nn—nn will be the node pair (for example, &501—the node pair is 01).
2	Calculate the checkpoint volume size requirement (see Table 7-9 in the Engineer's Reference Manual for the procedure to make this calculation).
3	Calculate the number of files (see Table 3-14 Line 51 of Network Form Instructions for the procedure to make this calculation).

Configure checkpoint volume on the HM**Table 2-21 Configuring HM Checkpoint Volume**

Step	Action
1	From the Engineering Personality Main Menu, select VOLUME CONFIGURATION .
2	Select the HM node pair that contains the AM checkpoints (determined in Step 1 of the previous procedure).
3	Select CHECKPOINT .
4	If you are adding a new node, move to the first blank row of entry boxes and add the Node Number, Volume Size (from Step 2 of the previous procedure), and the Number of Files (from Step 3 of the previous procedure).
5	If you are upgrading an existing node, find the entries for that node and enter the new values for Volume Size and Number of Files.
6	Press [CTL][F1] to check and [CTL][F2] to install the configuration.
7	Backup, initialize, and restore the HM (backup and restore are covered in subsection 2.4 of the Engineer's Reference Manual, and initialization is covered in Tasks 14 and 15 in the System Startup Guide).

Load AM personality with null checkpoint

This procedure loads the standard AM personality from the HM together with a null checkpoint from removable media.

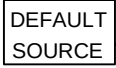
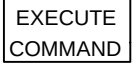
Table 2-22 Loading AM Personality and Null Checkpoint

Step	Action
1	Insert the floppy labeled “Startup AM Checkpoint” Volume ID &AMC, or the cartridge labeled &C6, in a removable media drive (this is the AM null checkpoint).
2	Press the [SYS STATS] button.
3	Select APPLICATION MODULES .
4	Select the desired AM node.
5	Select LOAD DUMP .
6	Select MANUAL LOAD .
7	Select COLD LOAD .
8	For PGM SOURCE, select DEFAULT SOURCE .
9	Select EXECUTE COMMAND .
10	For DATA SOURCE, select ALTERNATE SOURCE .
11	Select the drive that has the floppy or cartridge containing the null checkpoint.
12	Select EXECUTE COMMAND .
13	Press Enter to execute.
14	When the load is complete and the node status is OK, continue with the next procedure without changing screens.

Checkpoint the AM

This procedure creates a checkpoint for the new AM node on the HM.

Table 2-23 Checkpoint the New AM

Step	Action
1	Select  .
2	Select  .
3	Select  .
4	The STATUS column on the AM display shows SAVE.
5	When the save is complete and the AM status shows OK, continue with the next procedure without changing screens.

Shutdown node

You must shut down the AM (which is now loaded with standard AM personality) in preparation for the load of the A^XM personality.

Table 2-24 Shutting Down the New AM

Step	Action
1	Select  .
2	Press Enter to execute.
3	Repeat Steps 1 and 2 to do a second shutdown.

Reload other nodes

If your A^XM is a new node, you must shut down and reload all US and U^XS nodes from which you will access the A^XM. This will make the A^XM visible to those nodes.

Load the AM from the coprocessor hard drive

From now on you will load the A^xM node personality from the coprocessor hard drive using the AUTOLOAD NET function. This personality contains the functionality to communicate with the coprocessor.

Go to subsection 3.4 of this publication, Loading the AM Personality, read the material there, and perform the procedure to load the AM.

Note: If you loaded the AM prior to configuring networking (set_parms), you must reload the node.

3. Application Module^x Operations

3.1 Overview

What will be covered

This section covers the basic functions that are required to operate an A^xM. The following topics will be covered:

- Bootloading the Coprocessor with HP-UX

- Configuring AM External Load Modules

- Loading the AM Personality

- Shutting Down HP-UX

3.2 Bootloading the Coprocessor with HP-UX

Two methods will be covered

There are two basic ways to cause a bootload of the coprocessor when it is shut down and halted:

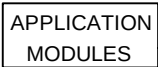
Cycle power

Use the WSI Restart Schematic at a US, U^XS or GUS

Cycle power

Note: The following procedure will kill the AM and require that you reload its personality. If this is undesirable, use the WSI Restart Schematic method instead of cycling power.

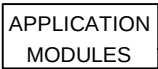
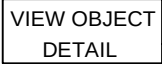
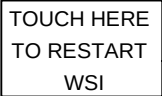
Table 3-1 Bootloading HP-UX by Cycling Power

Step	Action
1	If the AM has personality loaded, perform the following from a US, U^XS or GUS: Press the [SYS STATS] button Select  Select the desired AM node Select  Press Enter Repeat these steps a second time if you wish to avoid communications error messages in your logs.
2	When the AM shutdown is complete, turn off the power to the Five-Slot or Ten-Slot Module.
3	Wait approximately 10 seconds and then turn power back on.
4	HP-UX will boot automatically. If you are not monitoring the boot on a console terminal, you can wait for the yellow indicator on the coprocessor board to turn on, indicating that the boot is complete.
5	Load the AM-side personality.

WSI Restart Schematic

Assuming that the coprocessor is down, the following procedure will reset the coprocessor causing a reboot of HP-UX without requiring a reload of the AM personality.

Table 3-2 Bootloading the Coprocessor from a Schematic

Step	Action
1	Perform the following from a US, U ^X S or GUS: Press the [SYS STATS] button.
2	Select  .
3	Select the desired A ^X M node.
4	Select  .
5	Page forward to Page 2. The coprocessor status will be given; for example: WS INTERFACE WARNING
6	Select this status and then select  .
7	Assuming that the status is WARNING, select the  target and then press Enter.

3.3 Configuring AM External Load Modules

Attention

The discussion included here is useful regardless of whether you are adding a new A^XM node or upgrading an existing AM to an A^XM.

Use the procedures here if you are upgrading an existing AM to an A^XM.

If you are adding a new A^XM node on the LCN, it is more straightforward to configure the external load modules in the process of creating the new node. If your A^XM is a new node, perform the initial setup procedures in Section 2 and the procedures in subsection 2.4, “Creating a New Node,” which include configuring external load modules while configuring the node.

Overview

Prior to loading the AM personality, which is the next topic, you should make the NCF changes to configure any external load modules that you want included in the personality load. Refer to the *CL/AM Reference Manual*, Appendices C-I, for the various modules that can be loaded

Modules covered here

Two modules are specific to the Application Module^X, and these will be covered briefly here. They are:

AMCL06_2—Application Module^X CL Runtime Extensions (covered in Appendix I of the *CL/AM Reference Manual*)

XACCES—A module which, if loaded, enables the READWRIT option in the global X-access switch, allowing non-CL-initiated X-side applications to write LCN data if the global X-access switch is set to the READWRIT state (additional information is available in the *Application Module^X User Guide*.)

Function of AMCL06_2

This external load module contains the CL runtime extensions that allow an AM/CL program (running in an A^XM) to:

- Initiate, activate, and terminate X-side applications
- Monitor the status of the application initiation queue
- Change the state of the global X-access switch

This module is required and must be loaded or the AM-side of the A^XM will crash during startup.

Discussion

The global X-access switch controls when an X-side application can write LCN data. It is implemented by a Processor Status Data Point (PSDP) parameter called `$XACCESS` which has three possible states:

`READONLY`—The X-side can read but cannot write LCN data (default)

`RW_LCN_I`—The X-side can read LCN data, but can write LCN data only from applications that are initiated by CL on the LCN-side

`READWRIT`—The X-side can read and write LCN data from CL-initiated and non-CL-initiated applications (this state cannot be set by any means unless the external load module `XACCES` is loaded)

The state of the global X-access switch can be changed:

From the keyboard with Engineer access

From an LCN-side CL program (using a call available in `AMCL06_2`—the global X-access switch cannot be changed by a normal CL write to a parameter).

An X-side application cannot change the global X-access switch; however, a configuration file `xaccess.cfg` on the X-side provides the restart value for the switch. The `xaccess.cfg` file is modifiable by the tool `xaccess` if the user has execute permission on the tool and read/write permission on the configuration file.

In all cases, the `READWRIT` state cannot be set unless the `XACCES` external load module is loaded in the AM-side.

Attention

`$XACCESS` is not a checkpointed parameter. When the A^xM node personality is loaded, the restart state of `$XACCESS` is determined solely by a value encrypted in the X-side file `xaccess.cfg`. (The X-side must be running for the A^xM node personality to load.) If the X-side is restarted while the node personality is running, state of `$XACCESS` is not affected—it will remain in its current state.

Function of XACCES

XACCES, by its presence or absence, allows or disallows writes of LCN data from non-CL-initiated X-side applications. This is accomplished as follows:

If XACCES is loaded, the READWRIT state of the global X-access switch can be set by an Engineer from the keyboard, or by a CL call. When the READWRIT state is set, non-CL-initiated applications are allowed to write LCN data.

If XACCES is not loaded, the READWRIT state cannot be set, and therefore non-CL-initiated applications cannot write LCN data.

Installation on History Module**Table 3-3 Installing External Load Modules on the HM**

Step	Action
1	Using the Command Processor of the Engineering or Universal personality, determine if the directories &CUS and &CLX exist on the History Module: LSV NET (lists volumes on the network)
2	If the directories exist, verify that the directory &CUS contains the files AMCL06_2.LO and XACCES.LO, and that the directory &CLX contains the file AMCL06_2.SF: LS NET>&CUS LS NET>&CLX
3	If either or both directories do not exist, use the Create Directory command to create the missing directory or directories on the History Module: CD NET>vol>&CUS (vol is an existing volume on the HM) CD NET>vol>&CLX
4	If you had to create the directories or if they did not contain the required files, copy the files from the Application Module Personality cartridge (&C3)(x = the removable media drive number): CP \$Fx>&CUS>AMCL06_2.LO NET>CUS>AMCL06_2.LO CP \$Fx>&CUS>XACCES.LO NET>CUS>XACCES.LO CP \$Fx>&CLX>AMCL06_2.SF NET>CLX>AMCL06_2.SF

Configure the NCF

Table 3-4 Configuring External Load Modules in the NCF

Step	Action
1	At a US, U ^x S or GUS, install an NCF backup floppy or cartridge.
2	From the Engineering Main Menu, select SUPPORT UTILITIES .
3	Select MODIFY VOLUME PATHS .
4	Set the NCF Backup Path to the NCF backup media pathname: \$Fx>&ASY> (x = drive number)
5	Press Enter.
6	From the Engineering Personality Main Menu, select LCN NODES .
7	A display will appear with a target for each node possible on the LCN. Verify that the configuration is in the ON-LINE mode in the upper right corner of the display. Select the target for the node number of the A ^x M that you wish to configure (you may need to use the Page Forward key to display the target for the desired A ^x M).
8	Select the MODIFY NODE target.
9	Page forward to Page 3 and in the first box in the “NAME” column, enter AMCL06_2 (this module is required). Move to the adjacent box in the “PERS” column and add the personality type AM0. If you are going to install XACCES, enter its name, XACCES, and personality type, AM0. If you are installing any other optional external load modules, you may enter their names and personality type AM0 in exactly the same manner at this time.
10	Press Enter.
11	Press [CTL][F1] to check the new NCF, and press [CTL][F2] to install the new NCF.
12	Press Enter to confirm the install.
13	Return to the Engineering Personality Main Menu. A message will appear at the bottom of the display indicating the successful completion of the NCF installation.
14	Load the A ^x M node personality. The load will include AMCL06_2 (and XACCES, if configured).

3.4 Loading the AM Personality

Attention

If you are creating a new node on your LCN (as opposed to upgrading an existing AM to an A^XM), you must load standard AM personality from the HM together with a null checkpoint from removable media, then checkpoint the AM, and then shut it down. These steps must be performed before loading the personality as covered here.

If your A^XM is a new node, first perform the initial setup procedures in Section 2, including the procedures in subsection 2.4, that cover the creation of a null checkpoint on the HM.

Overview

You must use the AUTOLOAD NET function to load the A^XM personality in the LCN-side of an A^XM. The AUTOLOAD LOCAL and the MANUAL load functions cause the AM-side to load the standard AM software which will not interact with the coprocessor.

Attention

You must configure networking and reboot the coprocessor (see subsection 2.3) prior to loading the AM personality. If you loaded the AM prior to configuring networking and rebooting the coprocessor, you must reload the AM after these procedures are performed.

Coprocessor Status

The coprocessor must be up and running HP-UX in order for the AM-side to complete the process of loading the AM personality. If you initiate the load before the coprocessor completes its bootload, the standard AM personality will be loaded.

In the procedure that follows (Table 3-5), if the node status does not transition to READY or OK within approximately five minutes, HP-UX is probably not functional. Check the status detail as described in Table 3-5. If the WS INTERFACE status is WARNING, it indicates that the AM is not able to communicate with the coprocessor or that a licensing problem exists.

If the AM is not loaded, and the coprocessor is down, and you must load AM software to begin process control applications that can run independent of the coprocessor, you can do a MANUAL load. This will load the standard AM personality from the History Module. When the coprocessor is operational again, you will have to shut down and reload the node using AUTOLOAD NET to bring in the personality from the coprocessor hard drive. Then the AM-side and X-side will be able to communicate with each other.

Performing the load

Perform the following procedure from a US, U^xS or GUS:

Table 3-5 Procedure to Load AM Personality

Step	Action
1	Press the [SYS STATS] button.
2	Select APPLICATION MODULES .
3	Select the desired AM node.
4	Select LOAD/ DUMP .
5	Select AUTOLOAD NET .
6	The display will show the current default startup mode that is configured in the NCF (COLD, WARM, HOT, or NO POINT PROCESS). This is the startup mode that will be used unless you decide to override it. If you are satisfied with this mode, go to Step 8.
7	If you wish to override the default startup mode, select OVERRIDE DEFAULT. The targets HOT LOAD WARM LOAD COLD LOAD and NO POINT PROCESS will appear. Select the desired mode. This will override the NCF default startup mode, but only for this load.
8	Press Enter.
9	If you observe the load from the US, U^xS or GUS display, you will see the STATUS go through the following states: POWER ON QUALIFIED LOCAL LOAD NET LOAD READY WARNING—if it remains in this state for a long period of time, check the coprocessor status (WS INTERFACE on page 2 of the Status Detail display) OK You will also see the TYPE change from AM to A^xM.

For more information

For more information, refer to the *A^xM Troubleshooting* manual. It contains a more detailed description of the load process.

3.5 Shutting Down HP-UX

CAUTION

Never stop (shut down) HP-UX by using the power switch. This practice can corrupt the file system.

Attention

If you do a complete shutdown and halt of HP-UX, be aware that you will have to power down the node (killing the AM-side) in order to reboot HP-UX. If you only need to go to single-user mode to perform system administration activities, use the shutdown command without the halt (-h) option and then use the reboot command when you have completed your administration tasks (see the procedure in “Partial shutdown” below).

Who can shut down HP-UX

Normally, only the system administrator or a designated superuser should shut down the system. Shutdown authority is determined by the contents of the file `/etc/shutdown.allow`. You can designate which users are authorized to run shutdown by listing those users in the file. If the file is empty, only the superuser has shutdown authority. If the superuser login (usually root) is not listed in the file and the file is not empty, the superuser will not have shutdown authority. If the file is not empty, only those users listed will have shutdown authority.

Partial vs. total

Two procedures are described:

Partial shutdown—takes the system to single user mode, from which it can be subsequently rebooted without cycling power.

Total shutdown—causes HP-UX to stop. Reloading requires using the Restart Schematic (See Table 3-2, Bootloading the Coprocessor from a Schematic) or turning power off and back on.

Use the partial shutdown procedure whenever possible as an alternative to a total shutdown. It will put the system in single-user mode, shutting down all processes that are not necessary for the single-user. From this mode, you can perform system administration tasks that would be inappropriate or risky if the system were in full operation (for example, a hard disk backup). The system can be rebooted later with the reboot command and you will not need to cycle power or use the Restart Schematic.

Note: While in the single-user mode, you can only access the system through the console terminal. Refer to the Honeywell publication *Application Module^x Service* for details on how to connect a console terminal.

Partial Shutdown

Table 3-6 Partial Shutdown of HP-UX

Step	Action
1	Log on as root or use the su command to assume the root identity (the root password is required).
2	Change to the root directory if not already there: <code>cd /</code>
3	Use the shutdown command: <code>shutdown n</code> where n is the grace period time in seconds—the time delay until the system actually shuts down. Logged on users are warned of the impending shutdown by messages. If no grace period is specified, the default is 60 seconds. Using a value of zero results in the system shutting down immediately.
4	HP-UX will go to the single-user mode. When you have completed your administration tasks, use the reboot command and the system will start up: <code>reboot</code>

Total shutdown

Note: This procedure will require you to turn power off and back on to reboot HP-UX, which will kill the AM-side and require reload of its personality.

Table 3-7 Total Shutdown of HP-UX

Step	Action
1	Log on as root, or use the su command to assume the root identity (the root password is required).
2	Change to the root directory if not already there: <code>cd /</code>
3	Use the shutdown command: <code>shutdown -h n</code> where n is the grace period time in seconds—the time delay until the system actually shuts down. The -h option causes the system to halt at the end of the shutdown process. In order to reboot, you will have to cycle power. Logged on users are warned of the impending shutdown by messages. If no grace period is specified, the default is 60 seconds. Using a value of zero results in the system shutting down immediately.

For more information

You can customize the command line shutdown process. For more information, refer to Chapter 3 of the Hewlett-Packard manual *System Administration Tasks*, which covers starting and stopping HP-UX.

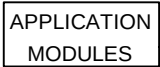
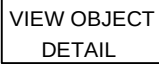
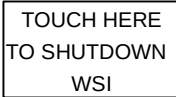
Using SAM for shutdown

You can use SAM to do the full (halt) shutdown or partial (single-user) shutdown. Using the procedure in subsection 4.1, invoke SAM and then select Routine Tasks, and then select Shutdown.

Shutdown from a US, U^xS or GUS schematic

You can shut down the coprocessor from a system schematic. Use the procedure in the following table to access the schematic.

Table 3-8 Shutting Down the Coprocessor from a Schematic

Step	Action
1	Perform the following from a US, U ^x S or GUS: Press the [SYS STATS] button.
2	Select 
3	Select the desired A ^x M node.
4	Select 
5	Page forward to Page 2. The coprocessor status will be given; for example: WS INTERFACE OK
6	Select this status and then select 
7	Assuming that the status is OK, select the  target and press Enter

4. Basic System Administration Tasks

4.1 Starting a System Administration Session

Starting SAM

The procedures in Section 4 assume that you are at the System Administration Station—a U^XS or workstation on the PIN, and that you are logged into the System Administration Station as root. Use the following procedure each time you start SAM.

Table 4-1 Starting SAM

Step	Action
1	Log on the System Administration Station as root and enter the root password.
2	When the prompt <code>TERM=(vt100)</code> appears, enter: <code>hpterm</code> and then enter: <code>export TERM</code>
3	Enter the command: <code>/usr/bin/X11/xhost +axmname</code> where <code>axmname</code> is the Internet hostname for the A ^X M NOTE: the third term in the command above is capital X followed by the number eleven.
4	Enter the command: <code>rlogin axmname</code> where <code>axmname</code> is the Internet hostname for the A ^X M.
5	Enter the root password.
6	Enter the commands: <code>TERM=hpterm</code> <code>export DISPLAY=hostname:0.0</code> <code>clear</code> where <code>hostname</code> is the System Administration Station hostname.
7	Enter the command: <code>sam</code>
8	On the menu bar, pull down View and select By Properties.

Running SAM from the console device

You can run SAM from the A^XM console terminal, but you will have to use the vt100 TERM environment. SAM is not as friendly to use in this environment. You must use the tab and arrow keys to navigate, instead of using the mouse. SAM displays on-screen directions when it is started in the vt100 TERM environment.

Running SAM from a PC or GUS using telnet

You can run SAM from a PC with Windows 95, 98, or NT (including GUS) using the telnet facility. As is the case when using a local console terminal, you will be operating SAM in the character mode and will have to navigate menus with the arrow and tab keys.

4.2 Backups and Restores

Backup procedure with remote DAT

Use the following procedure if your A^xM does not have a locally-connected DAT drive.

Note: The System Administration Station must have a properly configured `/ . r h o s t s` file—see subsection 2.2, Table 2-3.

Attention

To increase your chances of creating a good backup tape, we recommend that you use a DAT cleaning cartridge to thoroughly clean your DAT drive before creating backup tapes. We also recommend that you make AT LEAST two backup tapes.

Table 4-2 A^xM Backup with Remote DAT

Step	Action
1	Place backup tape in the system administration device DAT drive.
2	Log into the A ^x M and invoke SAM (see Table 4-1, Starting SAM).
3	Select Backup and Recovery from SAM.
4	Select Interactive Backup and Recovery.
5	A message will be displayed that begins with the sentence “No valid backup devices were found during the hardware scan.” Select OK.
6	On the menu bar, pull down the Actions menu.
7	Select Use Remote Backup Device.
8	Select Backup Files Interactively.
9	Select Specify Remote Backup Device.
10	Select the Remote System Name entry box and enter the Internet (IP) hostname of the System Administration Station.
11	Select the device File entry box and enter the filename of the tape device file: <code>/dev/rmt/0m</code> (zero, not the letter “oh”) NOTE: This is the default device file—if you have changed the default device file, enter the correct filename.

Table 4-2 A^XM Backup with Remote DAT

Step	Action
12	Select OK
13	Select Select Backup Scope
14	The default backup scope is Specified Files. If you wish to select or exclude specific files, leave the backup scope as is and enter each desired filename in the “File Name” box and Select Add after each name is entered. If you wish to back up the entire system, change the backup scope to: Local File Systems Only (no NFS), or All File Systems (includes NFS) depending on whether or not you want to include locally mounted network file systems. You can exclude files or directories as needed.
15	Select OK.
16	Select Set Additional Parameters.
17	Select Create Index Log if you want to create a log (this is the default).
18	Select OK.
19	Select OK.
20	When prompted “Do you want to proceed”, select Yes.
21	When prompted again “Do you want to proceed”, select Yes.
22	When the process has completed, press Enter to continue.
23	On the menu bar, pull down the File menu and select Exit.
24	On the menu bar, pull down the File menu and select Exit SAM.

Backup procedure with local DAT

Use the following procedure if your A^XM has a locally-connected DAT drive.

Table 4-3 A^XM Backup with Local DAT

Step	Action
1	Place backup tape in the A ^X M DAT drive.
2	Log into the A ^X M and invoke SAM.
3	Select Backup and Recovery from SAM.
4	Select Interactive Backup and Recovery.
5	Select the Local DAT Drive.
6	On the menu bar, pull down the Actions menu.
7	Select Back Up Files Interactively.
8	Select Select Backup Scope.
9	The default backup scope is Specified Files. If you wish to select or exclude specific files, leave the backup scope as is and enter each desired filename in the "File Name" box and select Add after each name is entered. If you wish to back up the entire system, change the backup scope to: Local File Systems Only (no NFS), or All File Systems (includes NFS) depending on whether or not you want to include locally mounted network file systems. You can exclude files or directories as needed.
10	Use the Tab key to go to OK and then press Enter.
11	Select Set Additional Parameters, if needed. Consider selecting "Overwrite newer files" to make sure you restore all the files off the backup.
12	Select Create Index Log if you want to create a log (this is the default).
13	Use the Tab key to go to OK and then press Enter.
14	Use the Tab key to go to OK and then press Enter.
15	When prompted "Do you want to proceed," select Yes.

Table 4-3 A^xM Backup with Local DAT

Step	Action
16	If prompted again, “Do you want to proceed,” select Yes.
17	When the process has completed, press Enter to continue. A Confirmation window appears.
18	On the menu bar, pull down the File menu and select Exit.
19	On the menu bar, pull down the File menu and select Exit SAM.

Restore procedure with remote DAT

Use the following procedure if your A^xM does not have a locally-connected DAT drive.

Note: The System Administration Station must have a properly configured `/ .rhosts` file—see subsection 2.2, Table 2-3.

Table 4-4 A^xM Restore with Remote DAT

Step	Action
1	Place backup tape to be read in the system administration device DAT drive.
2	Log into the A ^x M and invoke SAM (see Table 4-1).
3	Select Backup and Recovery.
4	Select Interactive Backup and Restore.
5	A message will be displayed that begins with the sentence “No valid backup devices were found during the hardware scan.” Select OK.
6	On the menu bar, pull down the Actions menu.
7	Select Use Remote Backup Device.
8	Select Recover Files or Directories.
9	Select Specify Remote Backup Device.
10	Select the Remote System Name entry box and enter the Internet (IP) hostname of the System Administration Station.

Table 4-4 A^xM Restore with Remote DAT

Step	Action
11	Select the device File entry box and enter the filename of the tape device file: /dev/rmt/0m (zero, not the letter “oh”) NOTE: This is the default device file—if you have changed the default device file, enter the correct filename.
12	Select OK.
13	Select Select Recovery Scope.
14	The default backup scope is Selected Files. If you wish to restore specific files, leave the backup scope as is and enter each desired filename in the “File Name” box and select Add after each name is entered. If you wish to restore the entire system, change the backup scope to All Files on Media.
15	Select OK.
16	Select Set Additional Parameters.
17	Review the options—You may wish to select “Overwrite newer files” to make sure you restore all the files from the backup.
18	Select OK.
19	Select OK.
20	When prompted, “Do you want to proceed?” Select Yes.
21	When prompted again, “Do you want to proceed?” Select Yes.
22	When the restore has completed, press Enter to continue.
23	On the menu bar, pull down the File menu and select Exit.
24	On the menu bar, pull down the File menu again and select Exit SAM.

Restore Procedure with local DAT

Use the following procedure if your A^xM has a locally-connected DAT drive.

Table 4-5 A^xM Restore with Local DAT

Step	Action
1	Place backup tape to be read in the A ^x M DAT drive.
2	Log into the A ^x M and invoke SAM.
3	Select Backup and Recovery.
4	Select Interactive Backups and Restores.
5	Select the Local DAT Drive.
6	On the menu bar, pull down the Actions menu.
7	Select Recover Files or Directories.
8	Select Select Recovery Scope.
9	The default backup scope is Selected Files. If you wish to restore specific files, leave the backup scope as is and enter each desired filename in the “File Name” box, and Select Add after each name is entered. If you wish to restore the entire tape, change the backup scope to All Files on Media.
10	Select OK
11	Select Set Additional Parameters
12	Review the options—normally you will use the defaults unless you need to restore the files to a different path.
13	Select OK
14	Select OK
15	When prompted, “Do you want to proceed?” Select Yes
17	When the restore is complete, press Enter to continue.
18	On the menu bar, pull down the File menu and select Exit.
19	On the menu bar, pull down the File menu again and select Exit SAM.

Using Ignite-UX for System Backup and Recovery

Once you have completely upgraded your system to HP-UX 10.20 and performed all the necessary system configurations, you can use Ignite-UX and your local DAT drive to create a bootable system recovery tape. This tape will actually contain a compressed image of your system configurations, including customized ones, and a backup of all files on the root disk or volume group. Files that are on a non-root disk or in a non-root volume group are not backed up on this tape.

It is important that you:

- Create this tape after the installation and any customizations are complete.

- Create a new bootable recovery tape after any system configuration changes are made such as changing the logical volume sizes.

You can use the bootable recovery tape to restore a complete image of your root disk or volume group to the hard drive. Keep in mind that when you use your system recovery tape it will restore all previous configurations including the type of disk management system that was used. Also keep in mind that you will still need to restore any files that have been updated since the last bootable recovery tape was created.

This procedure requires that the following items be connected to the A^xM:

- Local DAT drive

Attention

To increase your chances of creating a good backup tape, we recommend that you use a DAT cleaning cartridge to thoroughly clean your DAT drive before creating backup tapes. We also recommend that you make AT LEAST two backup tapes.

Table 4-6 Creating a Bootable System Recovery Tape

Step	Action
1	Place backup tape in the A ^x M DAT drive.
2	Unmount all network file systems by entering the following command: <code>umount -F nfs -a</code>
3	Enter the command: <code>cd opt/ignite/bin</code>
4	Enter the command: <code>./make_recovery -A -v</code>
5	The following message appears: ***** HP-UX System Recovery Creating the recovery tape may take as long as 45 minutes.
6	The following message will appear when completed: System Recovery Tape Successfully Created

Table 4-7 Performing a System Recovery

Step	Action
1	Power down the system if it is not already powered down.
2	Power up the A ^x M.
3	Place the system recovery tape in the DAT drive and closely monitor the boot process on the console terminal. Be prepared to press the Esc key.
4	Press Esc to stop the boot process when you see the line: Press [Esc] to discontinue the Auto Boot process.

Table 4-7 Performing a System Recovery

Step	Action
5	This will bring up a main menu as follows: <pre> Auto Boot aborted. ==== MAIN MENU ===== Key Operation ----- 1 Boot From a Device 2 Path Configuration 3 Mode Configuration 4 Interactive Testing 5 Firmware Information 6 Hardware Information 7 System Configuration ... 77 Reset the System 88 Change Mode 99 Restart Auto Boot ----- Press Key, then press [Enter/Return] Select "Boot From a Device" by pressing 1 and then Enter. </pre>
6	The firmware scans for all potential boot devices available to the A^xM and presents them in the "BOOT FROM DEVICE" menu similar to the following example. Note that boot device names may be different on your system: <pre> Scanning for Boot devices. Please wait... ==== BOOT FROM DEVICE ===== Key Boot Device ----- 1 SCSI.6.0 SEAGATE ST34572N 2 SCSI.5.0 HP C2247-300 3 SCSI.2.0 HP HP35470A ... 0 Previous Menu 33 Effective ISL Mode [AUTOMATIC] 66 Rescan for Boot devices 77 Reset the System ----- To boot from a device, Press Key, then press Enter Select the appropriate tape device by pressing the Key number and then pressing Enter. </pre>

Table 4-7 Performing a System Recovery

Step	Action
7	The boot process should begin and the screen should be similar to the following example: Now booting. ISL Revision A.00.38 OCT 26, 1994 ISL booting hpux (;0):INSTALL Boot : tape(2/0/1.3.0.0.0.0.0;0):INSTALL 4999156 + 446464 + 361504 start 0x184268
8	DO NOT press return when you see the following screen message: WARNING: The configuration information calls for a non-interactive installation. Press <Return> Within 10 seconds to cancel batch-mode installation: Doing so will invoke the Interactive Recovery procedure and it will NOT AUTOMATICALLY restore many of your system parameters and configurations.
9	There will be additional screen messages including a list of directories and archives being restored. When the process is complete, which may take up to an hour, you will receive the following message: The system is ready.
10	Reboot the system.

Restoring Updated Files

After restoring your system using the Ignite-UX recovery tape, you will still need to update any of the files that have changed since the bootable recovery tape was created. Additionally, if there were ANY changes to the logical volume sizes since the tape was made, you will need to reset the volume sizes to reflect the most recent configurations before restoring updated files.

Table 4-8 Restoring File Backups

STEP	ACTION
1	If no changes were made to the logical volume sizes since the last bootable recovery tape was created, then go to Table 4-5, A ^x M Restore with Local DAT, and perform the procedures there. Otherwise, go to the next step.
2	Invoke SAM.
3	Go to Single User Mode by selecting Routine Tasks: Tab to Single User state Press return Select Immediate Shutdown Tab to OK Press return.
4	Restore the most recent LVM configurations by restoring the following entire directory: /etc/lvmconf
5	Enter the following command: /sbin/vgcfgrestore -n vg00 /dev/rdisk/c0t6d0
6	Enter the following command: /sbin/extendfs /dev/vg00/rlvol# Where # is the logical volume number. I.e. rlvol7 = /users for each logical volume that has changed.
7	Reboot the system.
8	Restore the latest full backup. Add incrementals if any.
9	Reboot the system.

4.3 Adding and Removing Users

Adding a user

Table 4-9 Adding a User

Step	Action
1	Log into the A ^x M and invoke SAM.
2	Select Accounts for Users and Groups
3	Select Users
4	On the menu bar, pull down the Actions menu.
5	Select Add
6	Enter the user's Login Name.
7	For User Identity (UID), use the default supplied by SAM.
8	Select the Home Directory data entry box—SAM will create a default path by concatenating “/home/” and the Login Name. Change /home /login_name to /users/login_name
9	Select the entry box for Primary Group Name and change the default supplied by SAM to: axm
10	Use defaults for Start-up Program and Login Environment, or make desired changes.
11	Enter optional information, if desired.
12	Select OK
13	Enter a password for the user, if desired. If you enter a password, you will be prompted to re-enter it.
14	Select OK
15	A window appears confirming that the user has been added. Select OK
16	On the menu bar, pull down the File menu and select Exit.
17	On the menu bar, pull down the File menu again and select Exit SAM.

Deleting a user

Table 4-10 Removing a User

Step	Action
1	Log into the A ^x M and invoke SAM.
2	Select Accounts for Users and Groups
3	Select Users
4	Select (highlight) the user you wish to remove.
5	On the menu bar, pull down the Actions menu.
6	Select Remove
7	Select the desired disposition of user's files and directories.
8	Select OK
9	When asked if you want to continue, Select Yes.
10	A window appears confirming that the user has been deleted. Select OK
11	On the menu bar, pull down the File menu and select Exit.
12	On the menu bar, pull down the File menu again and select Exit SAM.

4.4 Adjusting Swap Space

Overview

The Release 300 A^XM is shipped from the factory configured with 128 MB of coprocessor device swap space. If you are updating your operating system from HP-UX 9.05 to 10.20 by purchasing the 4 GB hard drive upgrade, this drive also comes configured with a default of 128 MB of device swap space. This is considered appropriate for a 32 MB coprocessor. If you ordered your A^XM with 64, 128, or 256 MB of coprocessor memory, or if you purchased and installed a memory upgrade to 64, 128, or 256 MB, you should increase the swap space. This procedure covers adding file system swap space. If you prefer to increase the amount of device swap space (which we recommend), refer to the HP documentation.

Recommended swap space

The Honeywell recommended guideline for swap space is 2.5 times the size of the coprocessor memory. The following table shows the amount of additional file system swap space that will bring the total swap up to the recommended amount.

Table 4-11 Recommended File System Swap

Coprocessor Memory (MB)	Recommended Additional Swap (MB)
64	32
128	192
256	512

Attention

The recommended additional file system swap is in addition to the 128 MB of device swap that is pre-configured by Honeywell.

Swap space procedure

Table 4-12 Configuring Additional File System Swap

Step	Action
1	Log into the A*M and invoke SAM.
2	Select Disks and File Systems.
3	Select Swap
4	From the menu bar, pull down the Actions menu.
5	Select Add File System Swap
6	Select the line that contains the name of the file system which needs the file system swap space configured.
7	Select the data entry box for Minimum Swap and enter the amount of file system swap to be configured (in MB).
8	Select the data entry box for Maximum Swap and enter the same value that you entered in the previous step.
9	Verify that the buttons for Now and At Every System Boot are depressed.
10	Select OK
11	You may get a note indicating that the swap space has been successfully added. Select OK
12	On the menu bar, pull down the File menu and select Exit.
13	On the menu bar, pull down the File menu again and select Exit SAM.

4.5 Changing the Size of Logical Volumes

Overview

LVM, which is a disk management subsystem, lets you allocate disk space according to the specific or projected sizes of your file systems or raw data. If your hard drive is configured with LVM, you can use SAM to change the volume sizes, and also to check the size of current volumes.

However, there may be times when you are unable to use SAM to change the size of logical volumes because you may have to unmount the filesystem in which SAM resides. Alternatively, you may also use the command line method to change logical volume sizes.

If you do change the size of your logical volumes, you should create an updated bootable system recovery tape so that if it becomes necessary to use the tape for system recovery, the most current logical volume sizes will be restored. See the Using Ignite-UX for System Backup and Recovery procedure in subsection 4.2 for more information on creating a recovery tape.

Changing logical volume sizes

Table 4-13 Changing Logical Volume Size Using SAM

Step	Action
1	Log into the A ^x M and invoke SAM.
2	Select Disks and File Systems
3	Select Disk Devices
4	Select the line that corresponds to the disk which you wish to change the size of.
5	From the menu bar, pull down the Actions menu.
6	Select Increase Size
7	Enter the desired new size of the volume in MB.
8	Select OK
9	When the process is complete, pull down the File menu and select Exit.
10	On the menu bar, pull down the File menu again and select Exit SAM.

Table 4-14 Changing Logical Volume Size Using the Command Line

Step	Action
1	Log into the A ^x M.
2	Determine the logical volume name (for example: /dev/vg00/rlvol3) of the filesystem you need to increase by entering the command: <code>bdf grep dev</code>
3	Go to single user mode by entering the command: <code>/sbin/shutdown now</code>
4	Unmount the filesystem by entering the command: <code>/sbin/umount [filesystem]</code> (for example: /sbin/umount/var)
5	Extend the logical volume size by entering the command: <code>/sbin/lvextend -L XXX [logical volume name]</code> XXX = size in megabytes you want the volume to be Logical volume name = name obtained in step 2
6	Extend the filesystem by entering the command: <code>/sbin/extendfs [raw logical device]</code> (for example: /dev/vg00/rlvol3)

4.6 Adding a Second Disk Drive

Install the physical drive

Refer to the *Application Module^X Service* manual for the procedure to install the new drive. Be sure to pin it properly. The second drive must be pinned for SCSI address 5. After the physical drive has been installed and the system powered up and bootloaded, perform the following procedure to add the drive to the software configuration.

Configuration procedure

Table 4-15 Configuring a Second Disk Drive

Step	Action
1	Log into the A ^X M and invoke SAM
2	Select Disks and File Systems
3	Select Disk Devices
4	Select the line that corresponds to the disk that you are adding—it will say “unused” in the Use column.
5	From the menu bar, pull down the Actions menu.
6	Select Add
7	In the Use Disk for box, accept the default value, File System.
8	In the Mount Directory box, enter a name of your new disk (the name must begin with a “/”), for example: /disk2
9	Select Modify Defaults to view the default options. The default options will normally be correct. If you are adding a new (blank) drive, the Create New File System should be selected (depressed). If you are adding a drive that already has files on it, deselect this option.
10	Select OK You will be returned to the Add a Hard Disk Drive menu.
11	If you get a confirmation window asking you if you want to proceed with creating a new file system, select YES.
12	Select OK NOTE: The time required to create the file system can vary from minutes to hours depending upon the size of the drive.
13	When the process is complete, pull down the File menu and select Exit.
14	On the menu bar, pull down the File menu again and select Exit SAM

4.7 Adding a CD-ROM

Install the drive

Install the CD-ROM drive using the upgrade kit instructions that accompanied the drive. Be sure to pin it properly. The *Application Module^X Service* manual also has information on the pinning and cabling of the CD-ROM drive. After the drive has been installed and the A^XM powered up and bootloaded, perform the following two procedures to add the drive to the software configuration.

Configuration procedure

Table 4-16 Configuring a CD-ROM Drive

Step	Action
1	Insert the HP-UX CD-ROM in the drive.
2	Log into the A ^X M and invoke SAM.
3	Select Disks and File Systems
4	Select CD-ROM, Floppy, and Hard Disks
5	Select the line: 2/0/1.2.0 unused.....SCSI CD-ROM drive
6	From the menu bar, pull down the Actions menu
7	Select Add a Hard Disk Drive
8	Select Set Disk Usage and Options
9	Select the Mount Directory input field
10	Enter the following: /cdrom
11	Select OK You will be returned to the Add a Hard Disk Drive menu.
12	When asked if the CD-ROM file system has a cdfs file system, Select Yes
13	Select OK The “unused” field (see Step 5) changes to “cdfs” when the process is complete.
14	On the menu bar, pull down the File menu and select Exit.
15	On the menu bar, pull down the File menu again and select Exit SAM.

Removing the CD-ROM disk

Before removing the CD-ROM disk from the drive, enter the command:

```
umount /cdrom
```

Then press the button on the front of the drive and the disk will eject automatically.

Inserting a CD-ROM disk

After inserting a CD-ROM disk in the drive, enter the command:

```
mount /cdrom
```

Accessing the CD-ROM

Use the following procedure to access the CD-ROM from the System Administration Station or from another X Windows device on the PIN.

Table 4-17 Accessing the A^xM CD-ROM

Step	Action
1	Log into the A ^x M and invoke SAM.
2	Enter the following command: lrom

4.8 Loading the Primary Drive

Overview

This procedure is used when you wish to replace your primary drive and you have already purchased the HP-UX 10.20 software upgrade kit and, as a result, do not wish to purchase a hard drive with the software already loaded on it. The procedure for upgrading the primary drive involves the following major steps:

- Verify that you have a system backup and recovery tapes

- Install new primary drive

- Configure new primary drive and restore system configurations and files

It is assumed that the drive being replaced already had HP-UX 10.20 installed and that you already have adequate backup tapes and an updated bootable recovery tape. See subsection 4.2, Backups and Restores, for information on creating backup and recovery tapes.

Requirements

This procedure requires that the following items be connected to the A^xM:

- Local console terminal

- Local DAT drive

Install New Primary Drive

CAUTION

In the following procedure, you will be handling disk drives. Be sure to use an ESD wrist strap and properly grounded work surface. Refer to the A^xM Service manual for assembly/disassembly instructions, drive locations, and pinning diagrams.

Table 4-18 Installing a New Primary Drive

Step	Action
1	When the system backup and the system recovery tapes are complete, shut down the AM side and shut down HP-UX (see subsection 3.5, Shutting Down HP-UX).
2	Power down the node.
3	Pin the new primary drive for same SCSI address as the previous drive.
4	Remove the old primary drive and install the new primary drive. Refer to the Application Module ^x Service manual for the procedure to install the new drive. Be sure to pin it properly.
5	Go to Table 4-7, System Recovery, and follow all the steps given to restore files to the new primary drive.

4.9 Upgrading the Secondary Drive

Overview

This procedure is used when you upgrade a secondary hard drive (for example, from 1.2 GB or 2 GB to 4 GB).

CAUTION

In the following procedure, you will be handling disk drives. Be sure to use an ESD wrist strap and properly grounded work surface. Refer to the A^xM Service manual for assembly/disassembly instructions, drive locations, and pinning diagrams.

Secondary drive upgrade procedure

Table 4-19 Upgrading the Secondary Drive

Step	Action
1	Log onto the A ^x M as root from the System Administration Station as described in Table 4-1, steps 1-6.
2	Determine the mount directory for the existing secondary drive by entering the command: <code>/etc/mount</code> The secondary mount directory is the first entry on the line that contains the string <code>/dev/dsk/c0t5d0</code> . The mount directory name begins with the “/” character. Record the name here: _____
3	Perform a backup of the secondary drive using SAM as covered in subsection 4.2 of this manual. Under “Select Backup Scope,” choose “Selected Files” and in the “Included” section, add the name of the mount directory from above (including the “/” character).
4	When the backup of the secondary drive is complete, shut down the AM side and shut down HP-UX (subsection 3.5).
5	Power down the node.
6	Pin the new secondary drive for SCSI address 5.
7	Remove the old secondary drive and install the new secondary drive.
8	Power on the node and log in as root.
9	Invoke SAM.
10	Select Disks and File Systems

Table 4-19 Upgrading the Secondary Drive

Step	Action
11	Select Disk Devices
12	Highlight the new drive that you added as the secondary drive (the Hardware Path will be identified as 2/0/1.5.0).
13	From the menu bar, pull down the Actions menu.
14	Select Add
15	Leave the default value, File System, in the Use Disk for box.
16	For Mount Directory, enter the name determined in step 2 of this procedure.
17	Select Modify Defaults and view the default options. The default options will normally be correct. If you are adding a new (blank) drive, the Create New File System should be selected (depressed). If you are adding a drive that already has files on it, deselect this option.
18	Select OK You will be returned to the Add a Hard Disk Drive menu.
19	If you get a confirmation window asking if you want to proceed with creating a new file system, select YES.
20	Select OK Creating a file system will take minutes to hours depending on the size of the drive.
21	When the process is complete, pull down the File menu and select Exit.
22	On the menu bar, pull down the File menu again and Select Exit SAM
23	Using the restore procedure in subsection 4.2, restore the data from the backup tape to the new secondary drive.
24	After the restore, do: cd <mount directory> ll where <mount directory> is the directory from step 2
25	The directories and files that were on the original secondary drive should be on the new drive.

5. Installing and Updating Software

5.1 Overview

Scope

This section covers procedures to update system software, install new application software, update existing application software, set up a software depot, and load software from a Tar-format tape. The procedures in subsections 5.2 and 5.3 are for Honeywell-supplied software distributed on Digital Data Storage (DDS) media in the Hewlett-Packard Software Distributor format. The procedure in subsection 5.5 is for tape in the Tar format.

Software Distributor format strategy

For these procedures, the A^xM must have access to a Digital Audio Tape (DAT) drive. If the A^xM does not have a local DAT drive, the software to be installed will need to be loaded onto a Software Depot Server that has a DAT drive using the *swcopy* utility. All or part of the software on the tape may be loaded into a directory that is the Software Depot. You may install or update software in more than one A^xM from the server using the utility *swinstall*. If you have a local DAT drive you can install directly from the tape using the *swinstall* command.

Tar-format strategy

Tar-format tapes can be loaded from a tape drive on the System Administration Station, or from a local DAT drive on the A^xM if present.

5.2 Loading HP-UX

Overview

This procedure is used to load the HP-UX operating system on the A^xM primary hard drive. Use this procedure when you need to:

- Install HP-UX 10.20 on a hard drive currently operating under HP-UX 9.05

- Recover from a hard drive crash

- Install HP-UX 10.20 on a new hard drive

Attention

If you are installing HP-UX 10.20 on a system currently operating under HP-UX 9.05, **DO NOT** perform this procedure unless you have a full backup of your system, especially the root drive. The cold install procedure will completely erase old data. By having a backup, you can revert to your 9.05 system if you encounter any difficulties with the procedure.

Requirements

This procedure requires that the following items be connected to the A^xM:

- Local console terminal

- Local DAT drive

Attention

In the following procedures, the key identified as Enter is labeled Return on many keyboards.

Installation procedure

Use the following procedure to install HP-UX on your primary hard drive.

Table 5-1 Loading the HP-UX base system

Step	Action
1	Shut down the AM node from a US, U*S or GUS.
2	Power down the system and connect the console terminal and DAT drive to the A*M (see the A*M Service Manual).
3	Power up the A*M and closely monitor the boot process on the console terminal. NOTE: Be prepared to press the Esc key.
4	Press Esc to stop the boot process when you see the line: Press [Esc] to discontinue the Auto Boot process.
5	Insert the HP-UX base operating system distribution tape in the DAT drive and press Enter.
6	This will bring up a main menu as follows: <pre> Auto Boot aborted. ==== MAIN MENU ===== Key Operation ----- 1 Boot From a Device 2 Path Configuration 3 Mode Configuration 4 Interactive Testing 5 Firmware Information 6 Hardware Information 7 System Configuration 77 Reset the System 88 Change Mode 99 Restart Auto Boot ----- Press Key, then press [Enter/Return] Select "Boot From a Device" by pressing 1 and then Enter.</pre>

Table 5-1 Loading the HP-UX base system

Step	Action
7	The firmware scans for all potential boot devices available to the A^xM and presents them in the "BOOT FROM DEVICE" menu similar to the following example. Note that boot device names may be different on your system: <pre> Scanning for Boot devices. Please wait... ==== BOOT FROM DEVICE ===== Key Boot Device ----- 1 SCSI.6.0 SEAGATE ST34572N 2 SCSI.5.0 HP C2247-300 3 SCSI.2.0 HP HP35470A ... 0 Previous Menu 33 Effective ISL Mode [AUTOMATIC] 66 Rescan for Boot devices 77 Reset the System ----- To boot from a device, Press Key, then press Enter Select the appropriate tape device by pressing the Key number and then pressing Enter.</pre>
8	The boot process should begin and the screen should be similar to the following example: <pre> Now booting. ISL Revision A.00.38 OCT 26, 1994 ISL booting hpux (;0):INSTALL Boot : tape(2/0/1.3.0.0.0.0.0;0):INSTALL 4999156 + 446464 + 361504 start 0x184268</pre> There will be additional messages that are not shown in the above example. The boot process will take 3 to 5 minutes.

Table 5-1 Loading the HP-UX base system

Step	Action
9	After the Install program has been booted from the install media, the install interface will be displayed as shown in the following example: Welcome to Ignite-UX! Use the <tab> key to navigate between fields, and arrow keys within fields. Use the <Return/enter> key to select an item. Use the <return> or <space bar> to pop-up a choices list. If the menus are not clear, select the "Help" item for more information. <pre style="text-align: center;"> [Install HP-UX] [Run a Recovery Shell] [Advanced Options] [Reboot] [Help]</pre> This Welcome screen is the first of several screens that will guide you through the installation process. Use the Tab key to highlight items on the screen and then press Enter to choose or activate the highlighted item. Use the Up or Down Arrow keys to navigate through the items in the menu. Select the "Install HP-UX" option and press Enter.
10	The next window is the User Interface and Media Options window. The default Source Location Options is "Media only installation." The default User Interface Options is "Guided Installation". Choose these defaults by tabbing to OK and pressing Enter.
11	The next window is the Install HP-UX wizard: Select an overall system configuration window. The default Configuration is HP-UX B.10.20 Default. Accept this default by pressing the Tab key to go to NEXT and then press Enter. NOTE: If you are using the Windows terminal emulator, you may see a one-column display that is not readable. If this happens, press Enter to go to the next screen.

Table 5-1 Loading the HP-UX base system

Step	Action
12	The next window is the Install HP-UX wizard: Select a system environment window. The default Environment is "image1: English HP_UX 10.20" Accept this default by pressing the Tab key to go to NEXT and then press Enter.
13	The next window is the Install HP-UX wizard: Select a root disk window. The default Root Disk is the primary hard disk in the A^xM. If you agree with the default selection, press the Tab key to go to NEXT and then press Enter. If there are two disks connected to the A^xM and you want the secondary disk to be the Root disk then press Enter on the highlighted/selected Root Disk. The "Root Disk Choices:" window will be displayed with a list of hard disks. Use the Up/Down Arrow key to navigate and press enter on the selected disk. Tab to NEXT and then press Enter to continue.
14	The next window is the Install HP-UX wizard: Specify the amount of root swap space window. The Install HP-UX wizard determines memory size and calculates a recommended device swap disk size that will be used unless you change it. Honeywell recommends that the root swap space be at least 2.5 times the size of the physical memory. If you wish to change the value calculated by the wizard: Press Enter to display the list of predefined selections of swap disk size. Use the Up/Down Arrow key to navigate, go to OK and press Enter. To continue to the next menu, press the Tab key to go to NEXT and then press Enter.
15	The next window is the Install HP-UX wizard: Select a file system type window. If your hard disk is greater than 2 GB, select the "Logical Volume Manager (LVM) with HFS." If your disk is 2 GB, you can select the above option, or you can select the whole disk option. The latter will give you a single partition encompassing the entire disk. NOTE: If your disk is 1 GB or less, you must use whole volume instead of LVM. To continue, press the Tab key to go to NEXT and then press Enter.
16	If you configured the Logical Volume Manager in the previous step, the next window is the Install HP-UX wizard: Specify root volume group disks window. Select the install program default and press the Tab key to go to NEXT and press Enter. If you did not configure the Logical Volume Manager in the previous step, the next window will be the language selection window, described in the next step.
17	The next window is the Install HP-UX wizard: Select a language(s) window. Select the default or browse the options by pressing Enter. To continue, press the Tab key to go to NEXT and then press Enter.

Table 5-1 Loading the HP-UX base system

Step	Action
18	The next window is the Install HP-UX wizard: Select Additional Software window. There is no additional software. Use the Tab key to go to NEXT and then press Enter.
19	The next window is the Install HP-UX wizard: Pre-install disk information window. Select the install program default and press the Tab key to go to NEXT and press Enter.
20	The next window is the Install HP-UX wizard: Pre-install check information window. A new window with WARNING message may appear to let you know that all the data in the disk will be destroyed. DO NOT continue if you have not done a full backup of your system - see subsection 4.2 of this manual. If you fully backed up your disk before starting this installation, press the Tab key to go to NEXT and then press Enter.
21	The next window is the Install HP-UX wizard: System Summary window. Select "Show Summary" and press Enter to display the General Summary and Hardware Summary tab. Use the Tab key to go to the selected window/topic and use the Up/Down Arrow key to scroll to other information. If your previous choices above are satisfactory, use the Tab key to go to FINISH and then press Enter. This phase of the installation will take about 30 to 45 minutes. The system will boot at the completion of this step.
22	The initial installation is complete when you see the message below: <pre> ----- Welcome to HP-UX! Before using your system, you will need to answer a few questions. The first question is whether you plan to use this system on a network. Answer "yes" if you have connected the system to a network and are ready to link with a network. Answer "no" if you: Plan to set up this system as a stand alone (no networking). Want to use the system now as a stand alone and connect to a network later. ----- Are you ready to link this system to a network? Press [y] for yes or [n] for no, then press Return. See the next step for information on the appropriate response.</pre>

Table 5-1 Loading the HP-UX base system

Step	Action
23	We suggest that you enter "Y" at this time as subsequent procedures use the configuration information provided in this step. If you answer "Y", you will be prompted for the following information: - System name and IP address - Root password - Time zone - Any subnets - Subnet mask - DNS server - NIS configuration - Gateway hostname and IP address If you answer "N", you will be prompted for this information only: - System name - Root password - Time zone NOTE: It is possible to perform configuration at a later time by restoring the appropriate configuration files from your backup tape or by running the script <code>/sbin/set_parms</code> with the proper argument (see the topic <code>set_parms</code> options in subsection 2.2 of this manual).
24	For either response, follow the instructions that appear on the screen. NOTE: skip configuration of the font server if this step comes up.
25	At the completion of this section, a congratulatory message appears. The system will automatically reboot and "HP-UX Start-up in progress" appears. This takes about 2 to 3 minutes. A column of functions (tasks) and a corresponding status column provides status indicators such as. OK, N/A or FAIL. If you configured networking, these should all be OK or N/A. If you did not configure networking, the following, items will indicate FAIL: - Configure LAN interfaces - NCS broker daemons - Start audio server daemon

5.3 Copying Software to a Depot

Overview

This procedure is used to copy or merge software selections from a software source to a target selection (software depot). This will allow you to copy software from the local DAT drive to the software depot. The copied software can then later be distributed to or installed on another system that is located on the same network, but that does not have an attached DAT drive. The procedure will be used when you need to:

- Set up a software depot

- Install non-operating software on a system without a local DAT drive

Attention

All systems on which the software will be loaded must already have HP-UX 10.20 installed. See section 5.2 of this procedure for information on installing HP-UX 10.20.

Requirements

This procedure requires that the following items be connected to the A^xM:

- Local DAT drive

Copy procedure

Use the following procedure to set up (register) a software depot and copy non-operating software from the local DAT tape to the primary hard drive.

Table 5-2 Using swcopy

Step	Action
1	Connect a local DAT tape if not already connected.
2	Log onto the system as root and enter the root password.
3	Install the appropriate software distribution tape in the DAT drive.
4	Enter the command: <code>/usr/sbin/swcopy</code> Press Enter.
5	A navigation display window appears explaining how to use the console terminal keys. Read the instructions if necessary. Press Enter to continue.

Table 5-2 Using swcopy

Step	Action
6	A SD Copy - Software Selection (hostname) (1) window appears.
7	A Select Target Depot Path (hostname) window appears with the default path set to <code>/var/spool/sw</code>. Press the Tab key to go to OK and then press Enter to accept this default.
8	NOTE: You may get a screen telling you that the Target Depot Path and the Source Depot Path cannot be the same. If this happens, press Enter to continue.
9	A Specify Source (hostname) window appears with the entry in the Source Depot Type highlighted (selected). If the Source Depot Type selected is not Local Tape, press Enter to display additional options. Use the Up/Down Arrow keys to select Local Tape.
10	In most cases, the system will already have determined the correct path for the DAT drive (Local Tape) to be <code>/dev/rmt/0m</code> Press the Tab key to go to OK and then press Enter.
11	From the Options menu, select Change Options. Uncheck the Enforce Dependency Check here.
12	An SD Copy - Software Selection (hostname) (1) window appears that lists the software products available on the source tape. Mark the software you wish to copy with a Yes by performing the following: Highlight the software product by pressing the space bar. Press the Tab key to go to the menu bar. Use the right arrow key to move to the Actions drop down menu and then press Enter. Use the down arrow key to select Mark for Copy from the Actions menu and then press Enter. The selected software should now be marked Yes.
13	An Error (hostname) window may appear if any of the software you selected has software dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies – it is just to let you know that they exist. Dependencies will be checked during software installation. Press Enter to return to the SD Copy – Software Selection window.

Table 5-2 Using swcopy

Step	Action
14	Repeat the above two steps for each software product that you wish to copy. When all the selections have been made, perform the following: Press the Tab key to go to the menu bar. Use the right arrow key to move to the Actions drop down menu and then press Enter. Use the down arrow key to select Copy Analysis from the Actions menu and then press Enter.
15	A Copy Analysis (hostname) (2) window appears displaying the Target, Status, and Products Scheduled information. When the analysis is complete, the Status indicates Ready or Ready with Errors (indicates that the program is unable to verify the software dependencies at this time). If you wish to look at the logfile, select Logfile and press Enter. A Logfile (hostname) (2) window appears. Select OK to return to the Copy Analysis window. Press the Tab key to go to OK and then press Enter.
16	A Confirmation (hostname) window appears. Select Yes and then press Enter.
17	A Copy Window (hostname) (3) window appears during the copy process displaying Target, Status, Percent Complete, Kbytes Installed, Time Left (minutes) and Copying Software information. The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Completed with Errors (indicates that the program is unable to verify the software dependencies).
18	Select Logfile and press Enter to display the copied software.
19	A Logfile (hostname) (2) window appears. Scroll through the log to determine if there were any errors during the copying process. Use the Tab key to go to OK and then press Enter to return to the Copy Window (hostname) (3) window.
20	Refer to the next section, Installing Other Software, when you are ready to install the copied software from the Software Depot source.
21	When you are ready to exit the copy program, perform the following: Press the Tab key to go to DONE and press Enter. Press the Tab key to go to the File selection of the menu bar and press Enter. Use the down arrow key to select Exit and then press Enter.

Note: If, at a later time, you no longer wish to use this system as a software depot, you can unregister the depot by using the *swremove* command. See the HP-UX man pages for further information on the *swremove* command.

5.4 Installing Other Software

Overview

If the AxM has a local DAT drive you may use “swinstall” to install the software directly from the tape. If the DAT drive resides on another computer, you must first copy the files from the tape to the other computer using the "swcopy" utility documented in the previous section to create a software depot. Then “swinstall” is executed on the AxM, and the source is set to the software depot on the other computer.

This procedure is used to load non-operating system on the A^xM. This software includes:

- A^xM Core Software

- TPS Process Network (LCN) A^xM node personality

- OpenDDA

- FORTTRAN and/or C++ compilers

- Honeywell and third-party applications

Requirements

This procedure requires that the following items are available to the A^xM:

- Local DAT drive, or a

- Software Depot (see section 5.3 for instructions on setting up a software depot and copying files to the depot)

Installation procedure

Use the following procedure to install non-operating software on your primary hard drive.

Table 5-3 Using swinstall

Step	Action
1	If you are installing software from a DAT tape: Connect the DAT drive if it is not already connected. Insert the appropriate software distribution tape into the DAT drive. Otherwise, go to the next step.
2	Log onto the system as root and enter the root password.
3	Enter the command: <code>/usr/sbin/swinstall</code> Press Enter.
4	A navigation window appears explaining the use of the console terminal keys. If you are not familiar with the use of these keys, read the instructions. Press Enter to continue.
5	A window titled SD Install - Software Selection (hostname) (1) appears.
6	The install software will scan the default source depot type and will display a Note (hostname) window with a message displayed and with the OK selection highlighted. Press Enter to continue.
7	A Specify Source (hostname) window appears with the entry in the Source Depot Type highlighted. Select the appropriate software source and path by performing the following: Press Enter to display additional options, if necessary - select local tape or software depot. Use the Up/Down Arrow keys to select your software source. Enter the correct path Press the Tab key to go to OK and then press Enter.
8	A Bundles and Products window appears that lists the software products available on the specified source. Mark the software you wish to install with a Yes by performing the following: Highlight the software product by pressing the space bar. Press the Tab key to go to the menu bar. Use the right arrow key to move to the Actions drop down menu and then press Enter. Use the down arrow key to select Mark for Install from the Actions menu and then press Enter. The selected software should now be marked Yes.

Table 5-3 Using swinstall

Step	Action
9	An Error (hostname) window may appear if any of the software you selected has software dependencies. This does not mean, however, that the target location will be unable to satisfy the dependencies - it is just a warning that dependencies do exist. Press Enter to return to the Bundles and Products window.
10	Repeat the above two steps for each software product that you wish to install. When all the selections have been made, perform the following: Press the Tab key to go to the menu bar. Use the right arrow key to move to the Actions drop down menu and then press Enter. Use the down arrow key to select Install (Analysis) from the Actions menu and then press Enter.
11	An Install Analysis (hostname) (2) window appears with Target, Status, and Products Scheduled information displayed. The install program is checking for adequate disk space and software dependencies. If all dependencies have been met and there is adequate disk space, the Status will indicate Ready. If the Status does not indicate ready, you will need to correct the problem before continuing. Press the Tab key to go to OK and then press Enter.
12	A Confirmation (hostname) window appears. Select Yes and press Enter.
13	Some software requires a system reboot after installation. If your system requires a reboot, another Confirmation (hostname) window will appear indicating that the system will be rebooted when installation is complete. If this occurs, select Yes and press Enter.
14	The Install Window (hostname) (3) window appears during the installation process displaying Target, Status, Percent Complete, Kbytes Installed, Time Left (minutes) and Loading Software information. The Percent Complete will advance to 100%. Do not, however, begin the next step until the Status indicates Completed or Ready.
15	Press the Tab key to go to Logfile and then press Enter to display all the loaded modules.
16	A Logfile (hostname) (2) window displays. Scroll through the log to determine if there were any errors during the installation. Press the Tab key to go to OK and then press Enter.

Table 5-3 Using swinstall

Step	Action
17	The system will return to the Install Window (hostname) (3) window. Press the Tab key to go to DONE and then press Enter.
18	If the software requires a reboot, a Note (hostname) window will appear informing you that the system will be rebooted. Press Enter to continue.
19	If the software does not require a reboot, you will get the " SD Install - Software Selection (hostname) (1) " window. Press the Tab key to go to the menu bar (File will be selected) and then press Enter. Select "Exit" from the pull-down menu and then press Enter.

5.5 Loading from a Tar-Format Tape

Overview

Tar-format tapes will generally be for installation or update of third-party application software. They will probably come with installation instructions; however, these instructions may need to be modified when using a remote tape drive.

Installation procedure with remote DAT

Use this procedure if your A^xM does not have a local DAT drive.

Table 5-4 Installing Software from a Tar-Format Tape (Remote DAT)

Step	Action
1	Insert the tape in the System Administration Station DAT drive.
2	Do an rlogin to the A ^x M as root from the System Administration Station.
3	Follow the instructions supplied with the tape relating to creating and/or changing to the directory where the software will be installed.
4	Enter the command: <code>remsh host dd if=/dev/rmt/0m bs=10k tar xf -</code> where host is the Internet hostname of the System Administration Station. Do not overlook the minus sign (-) at the end of the command—it is required.
5	It may take as long as 5 minutes for the tape to load. When the load is complete, the prompt will return to the screen.
6	Follow the instructions that were supplied with the tape for installation of the software.

Installation procedure with local DAT

Use this procedure if your A^xM has a local DAT drive.

Table 5-5 Installing Software from a Tar-Format Tape (Local DAT)

Step	Action
1	Insert the tape in the A ^x M DAT drive.
2	Do an rlogin to the A ^x M as root from the System Administration Station.
3	Follow the instructions supplied with the tape relating to creating and/or changing to the directory where the software will be installed.
4	Enter the command: tar -xv
5	It may take as long as 5 minutes for the tape to load. When the load completes, the prompt will return to the screen.
6	Follow the instructions that were supplied with the tape for installation of the software.

Index

A

- Adding a CD-ROM, 83
- Adding a Second Disk Drive, 82
- Adding and Removing Users, 76
- AMCL06_2
 - Function of, 52
- AUTOLOAD NET, 56
- A^xM Setup
 - Check access to the System Administration Station, 39
 - Configure other hosts, 35
 - Configure remote host permissions, 37
 - Console Log in Procedure, 34
 - End console session, 40
 - Overview, 29
 - Running set_parms, 34
 - Test networking, 40

B

- Backup procedure with local DAT, 67
- Backup procedure with remote DAT, 65
- Backups and Restores, 65
- Bootload HP-UX
 - Cycle power, 50
 - WSI Restart Schematic, 51
- Bootloading the Coprocessor with HP-UX, 50

C

- Configuring AM External Load Modules, 52
- Console connection
 - Dumb terminal procedure, 29
 - HP workstation procedure, 31
 - PC procedure, 32
 - U^xS procedure (R100), 30
 - U^xS procedure (R200), 30
- Creating a new node
 - Checkpoint the AM, 47
 - Configure checkpoint volume on the HM, 45
 - Configure the A^xM on the LCN, 43
 - Determine checkpoint volume requirements, 45
 - Installation on History Module, 42
 - Load AM personality with null checkpoint, 46
 - Load the AM from the coprocessor hard drive, 48
 - Overview, 41
 - Prepare NCF backup media, 42
 - Reload other nodes, 47
 - Shutdown node, 47
 - What will be covered, 41

D

- Deleting a user, 77
- Duties and Skills of a System Administrator, 16

E

- External load modules
 - Installation on History Module, 54

H

- Hewlett-Packard CD-ROM documentation, 18
- Hewlett-Packard web site, 18
- HP-UX Reference*
 - online version, 18
- HP-UX shutdown
 - For more information, 61
 - From a US, U^xS or GUS schematic, 62
 - Partial shutdown, 60
 - Partial vs. total, 59
 - Total shutdown, 61
 - Using SAM for shutdown, 61
 - Who can shut down, 59

I

- Ignite-UX, using for Backup and Recovery, 71
- Installing and Updating Software, 89
- Installing HP-UX on your primary hard drive, 91
- Installing Other Software, 100

L

- Levels of System Administration, 15
- Loading HP-UX, 90
- Loading the AM Personality, 56
- Logical Volume Size, using SAM to change, 80
- Logical Volume Size, using the command line to change, 81
- LVM, 80

M

- man command, 18
- man command limitations, 19

N

- NCF Backup Media, 42
- NCF, configuring external load modules, 55

Index

P

Primary Drive
 upgrading, 85
Primary Drive, upgrading, 85

R

Restore Procedure with local DAT, 70
Restore procedure with remote DAT, 68

S

SAM
 starting, 63
 using for shutdown, 61
Scope of this manual, 15
set_parms
 information required for, 24
 preparation for, 23
 Running on an HP-UX station, 27
 Running on the A^xM, 34
 Running on the system administration station, 23
set_parms options, 25
Shutting Down HP-UX. *See* HP-UX shutdown
Software Depot, setting up, 97
Starting SAM, 63
swap space, 78
Swap space procedure, 79
Swap space, adjusting, 78
System administration station
 Configure networking, 27
 Connect cables, 26
 Important information, 23
 Information required for set_parms, 24
 Preparation for set_parms, 23

Set up /etc/hosts and /.rhosts files, 28
Setup, 23

System Administration Station Network Configuration, 28
System administration strategy, 21

T

Tar-format strategy, 89
Tar-Format Tape, 104

U

Updating Software, 89
Upgrading the Primary Drive, 85
Upgrading the Secondary Drive, 87

V

Volume sizes, changing, 80
VT100 Emulation on a Hewlett-Packard Workstation, 31
VT100 Emulation on an R200 U^xS, 30

W

What is the Application Module^x?, 15
Where to Learn More
 Hewlett-Packard web site, 18
 HP CD-ROM manuals, 18
 Online version of the HP-UX Reference, 18

X

XACCES Discussion, 53
XACCES, function of, 54

FAX Transmittal

FAX No.: (602) 313-4842

TO: _____

Total FAX pages: _____
(including this page)

Reader Comments

Title of Document: **Application Module^x System Administration**

Document Number: **AX11-300**

Issue Date: **01/99**

Comments: _____

Recommendations: _____

FROM:

Name: _____ **Date:** _____

Title: _____

Company: _____

Address: _____

City: _____ **State:** _____ **ZIP:** _____

Telephone: _____ **FAX:** _____

FOR ADDITIONAL ASSISTANCE:

Write	Call
Honeywell Inc. Industrial Automation and Control Automation College 2820 West Kelton Lane Phoenix, AZ 85053-3028	Technical Assistance Center (TAC) 1-800-822-7673 (48 contiguous states except Arizona) 602-313-5558 (Arizona)

Honeywell

Industrial Automation and Control

Automation College
2820 W. Kelton Lane
Phoenix, AZ 85053-3028